

REPORT
on
**DAM BREAK ANALYSIS OF DEHRANG
DAM,
PANVEL, DIST. RAIGAD, MAHARASHTRA**

Submitted to
Panvel Municipal Corporation
Government of Maharashtra

Prepared by
**Centre for Climate Change and
Sustainability Studies, Shivaji University, Kolhapur**
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1. Introduction

1.1 General

Dehrang Dam is an earthen dam with a gated spillway, constructed in 1957 across the Gadhi River in the Raigad District of Maharashtra. Owned and operated by the Panvel Municipal Corporation (PMC), the dam serves as the primary source of raw water supply for the Panvel urban agglomeration and surrounding areas at a design rate of 150 LPCD. Situated approximately 18 km from Panvel, the dam is strategically located at Latitude 19 deg 01' 54" N and Longitude 73 deg 14' 32" E. The structure commands a catchment area of 27 km² over a hydrologically active portion of the Western Ghats foothills, characterised by steep slopes, dense forest cover, and extremely high annual rainfall ranging from 1,720 mm (minimum) to 4,417 mm (maximum), with an average of approximately 2,000 mm.

The dam has a total length of 335 m and is equipped with 11 Godbole-type Automatic Tilting Gates (each 8.70 m x 2.50 m), providing a total gated spillway length of 111 m. The Full Reservoir Level (FRL) and Maximum Water Level (MWL) are both set at 89.15 m MSL, with the Top Bund Level (TBL) at 91.65 m MSL providing a freeboard of 2.50 m above the FRL. The spillway gate sill level is 86.65 m MSL. Due to decades of siltation, the gross storage capacity at FRL has reduced from an original 11.01 MCum to approximately 2.70 MCum at present, as confirmed by a pre-monsoon sludge survey conducted by Yash Engineering Consultant Pvt. Ltd. in April 2025.

Downstream of the dam, the Gadhi River flows through a mixed landscape of agricultural land, rural settlements, and expanding peri-urban areas en route to the Panvel Creek and Ulwe River systems. The Panvel Municipal Corporation has therefore commissioned this Dam Break Analysis (DBA) in compliance with the Guidelines for Developing Emergency Action Plans for Dams (CWC, 2016) and the Dam Safety Act, 2021 and associated CDSO Guidelines (2018).

The technical assessment employs an integrated hydrologic-hydraulic modelling framework. Intensity-Duration-Frequency (IDF) curves are developed from 31 years of IMD daily rainfall data (1992-2024) using Gumbel EV-I frequency analysis. HEC-HMS 4.13 transforms the design storm into a reservoir inflow hydrograph, which serves as the upstream boundary condition for dam breach simulations in HEC-RAS 6.6 (2D unsteady flow). Two failure mechanisms - Overtopping and Internal Erosion (Piping) - are evaluated using Froehlich's (2008) empirical breach parameters.

1.2 Objective

The primary objective of this Dam Break Analysis is to establish a scientifically robust, model-driven safety assessment framework for the Dehrang reservoir under extreme hydrologic and structural loading conditions. Specific objectives are:

1. To develop IDF curves for the Panvel region using 31 years of IMD daily rainfall data and Gumbel EV-I frequency analysis, and to derive the 100-year 24-hour design storm using the Alternating Block Method (ABM).
2. To quantify the hydrologic inflow into the Dehrang reservoir using HEC-HMS 4.13, incorporating SCS Curve Number loss methodology and SCS Unit Hydrograph transformation.
3. To simulate dam breach behaviour under Overtopping and Internal Erosion (Piping) failure modes using a calibrated 2D unsteady flow model in HEC-RAS 6.6.
4. To generate breach hydrographs and downstream inundation mapping, including water surface elevations, maximum depths, flow velocities, wave arrival times, and hazard classifications.
5. To identify vulnerable downstream settlements, key infrastructure, and safe shelter locations to support preparation of an Emergency Action Plan (EAP).

1.3 Scope

The present study encompasses a complete hydrologic-hydraulic assessment of breach scenarios for Dehrang Dam. The scope includes:

1. Compilation and analysis of dam features including TBL, MWL, FRL, MDDL, spillway geometry, gate configuration, and storage characteristics from the 2025 pre-monsoon inspection report.
2. Statistical frequency analysis of 31-year annual maximum daily rainfall series using Gumbel EV-I and Lognormal distributions, with K-S and Chi-square goodness-of-fit tests.
3. Development of IDF curves and the 24-hour 100-year design storm using the IMD empirical reduction formula and the Alternating Block Method (ABM).
4. Hydrologic modelling in HEC-HMS 4.13 using the SCS-CN loss method and SCS Unit Hydrograph transform.
5. Two-dimensional hydraulic modelling in HEC-RAS 6.6, including 2D mesh construction, Copernicus 30m DEM integration, LULC-based Manning's roughness, and simulation of both overtopping and piping failure scenarios.
6. Breach parameter development using Froehlich (2008) empirical method, consistent with CDSO and CWC guidance.
7. Computation of flood depths, velocities, arrival times, hazard zones, and inundation extents for downstream areas.
8. Preparation of inundation maps and hazard/vulnerability maps identifying shelter points and evacuation routes.

1.4 Limitations

Despite rigorous modelling, certain inherent limitations apply to this study:

1. **DEM Resolution Constraints:** Terrain representation is based on the Copernicus DEM at 30 m horizontal resolution. Narrow drains, culverts, embankments, and micro-topographic relief may not be fully resolved.
2. **Assumption of Homogeneous Roughness:** Manning's n values are assigned based on LULC classes derived from satellite imagery. Intra-class variability and seasonal changes may influence local conveyance.
3. **Rainfall Data Limitations:** Two years of data (1997 and 2006) are missing from the 31-year record. The 2005 outlier event (473.5 mm) has been retained conservatively per CWC guidance.
4. **Simplified Representation of Hydraulic Structures:** Smaller field structures including local bunds, culverts, and road crossings may be approximated or excluded.
5. **Uncertainty in Breach Parameters:** Breach width, side slopes, and formation time are based on Froehlich (2008) empirical equations. Actual breach development may differ from modelled behaviour.
6. **Dynamic Land-Use Changes:** Ongoing urbanisation in the Panvel-Navi Mumbai corridor may alter future flood propagation characteristics.
7. **Model Dependency:** Results rely on HEC-HMS and HEC-RAS computation engines; inherent numerical tolerances and solver assumptions apply.

2. Description of the Dam / Reservoir System

Dehrang Dam is an earthen embankment structure with a gated spillway, built across the Gadhi River in 1957. The dam is situated in the Western Ghats foothills of the Raigad district and has been the primary water supply source for the Panvel Municipal Corporation for over six decades. The most recent pre-monsoon inspection was conducted by Yash Engineering Consultant Pvt. Ltd. in April 2025. The salient features of the dam are summarised in Table 1 below.

Sr. No.	Parameter	Value
1	Name of Project	Dehrang
2	Name of Dam	Dehrang Dam
3	Owner	Panvel Municipal Corporation
4	Nearest City	Panvel
5	River / Nalla	Gadhi River
6	Latitude	19 deg 01' 54" N
7	Longitude	73 deg 14' 32" E
8	Year of Construction	1957
9	Type of Dam	Earthen Dam with Gated Spillway
10	Top Bund Level (TBL)	91.65 m MSL
11	Full Reservoir Level / MWL (FRL)	89.15 m MSL
12	MDDL	80.77 m MSL
13	Spillway Crest Level (Gate Sill)	86.65 m MSL
14	Lowest Riverbed Level (RBL)	81.4 m MSL
15	Design Flood Lift for Spillway	2.50 m
16	Total Dam Length	335 m
17	Dam Width at Top	3.00 m (Revised: 3.30 m)
18	Berm Level (Revised)	82.50 m MSL
19	Berm Width (Revised)	6.50 m
20	Total Gate Length	111 m (11 Nos., 8.70 m x 2.50 m each)
21	Type of Gate	Godbole Type Automatic Tilting Gate
22	Catchment Area	27.00 km ²
23	Original Storage Capacity	11.01 MCum (Pre-siltation)
24	Revised Capacity (Post Gate Fixing)	3.507 MCum
25	Present Capacity (Post Siltation)	2.70 MCum (approx.)
26	Gross Storage at FRL (2025 Survey)	2,351.58 x 10 ³ m ³ (2.352 MCM)
27	Average Annual Rainfall	2,000 mm
28	Maximum Annual Rainfall	4,417 mm
29	Distance from Panvel	18 km

Table 1: Salient Features of Dehrang Dam

The gated spillway comprises 11 Godbole-type Automatic Tilting Gates, each 8.70 m wide and 2.50 m deep, providing a total controlled spillway length of 111 m. The gate sill level is 86.65 m MSL, providing a maximum water depth above the crest of 2.50 m at FRL (89.15 m MSL).

The reservoir has experienced significant siltation over its operational life. The original design capacity was 11.01 MCum; a detailed sludge survey in April 2025 confirmed the present gross storage at FRL to be $2,351.58 \times 10^3 \text{ m}^3$ (2.352 MCM), representing an 18.8% reduction from the pre-siltation capacity. The complete Elevation-Area-Capacity table is provided in Annexure I.

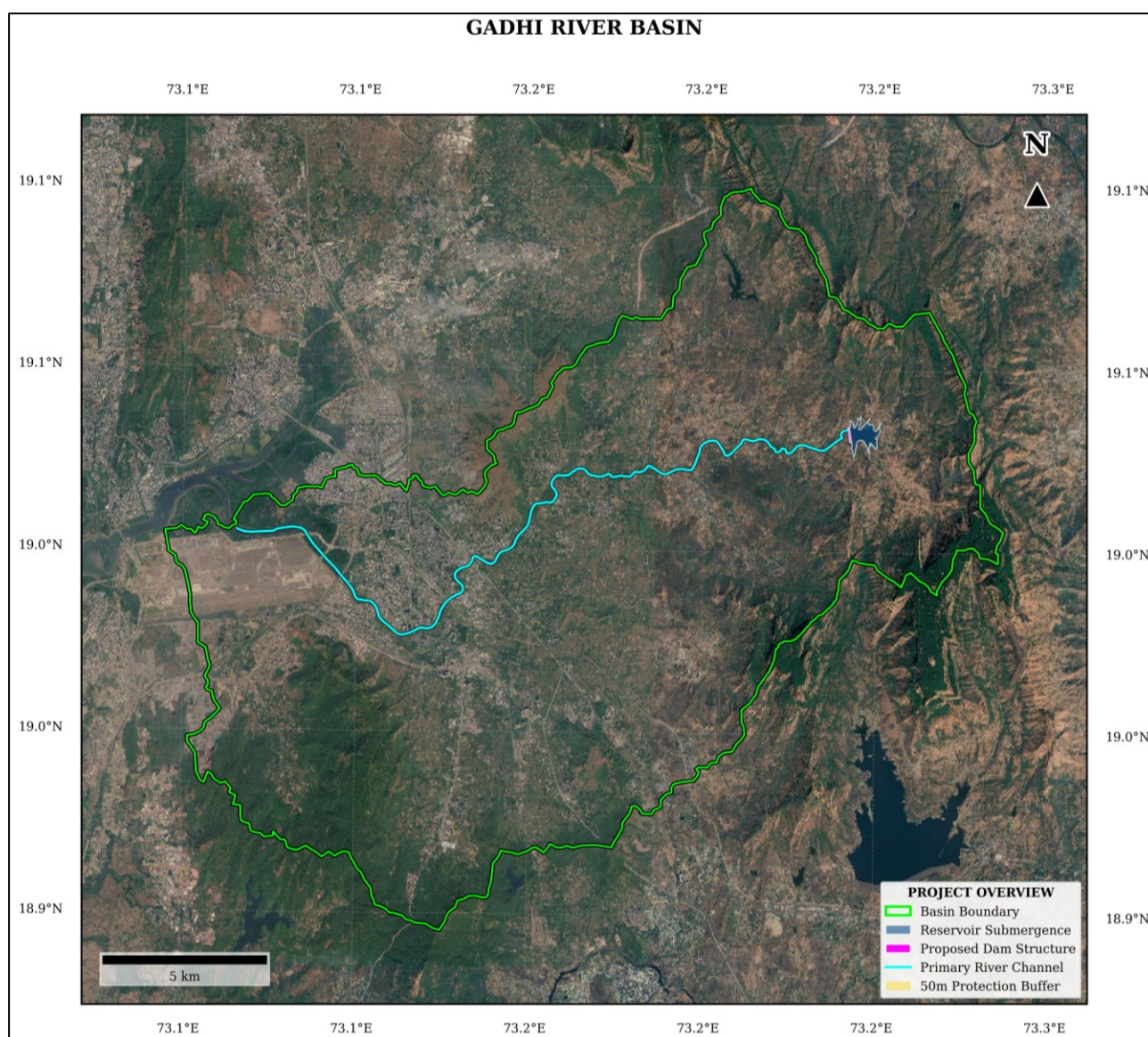


Figure 1.1: Study Area Comprehensive Map – Gadhi River Basin showing Basin Boundary, Reservoir, Primary River Channel and Downstream Flood Plain

3. IDF Analysis and Design Storm

3.1 Rainfall Data and Station Details

Annual maximum daily rainfall data spanning 31 years (1992-2024) was obtained from the India Meteorological Department (IMD) for the Panvel raingauge station [Station ID: 102187319], located at 18.9833 deg N, 73.1167 deg E. Two years (1997 and 2006) have missing data and are excluded, resulting in N = 31 observations. The series ranges from 118.6 mm (1995) to 473.5 mm (2005). Descriptive statistics: mean = 225.06 mm, S = 71.84 mm, Cv = 0.319, Cs = 1.357.

3.2 Frequency Analysis - Gumbel EV-I and Lognormal

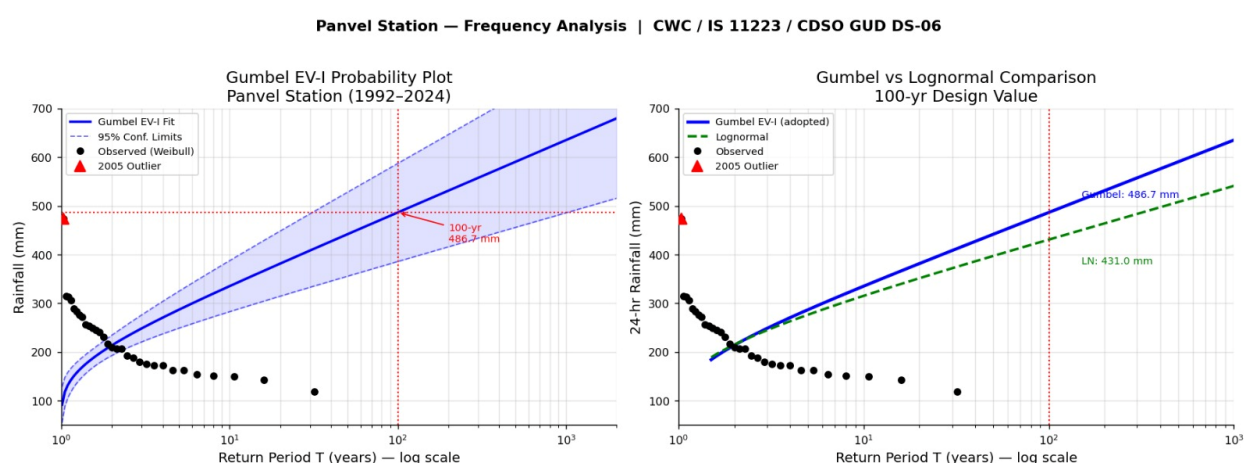


Figure 3.1: Gumbel EV-I Probability Plot and Lognormal Comparison – Panvel Station (1992–2024); 100-yr design rainfall = 486.7 mm

Frequency analysis was performed using Gumbel EV-I and Lognormal distributions as recommended by the CWC Manual on Flood Estimation (1989) and IS 11223. The Gumbel EV-I design rainfall uses the Chow formula $XT = \bar{x} + K \cdot S$, where $K = (y_T - y_n)/S_n$; $y_n = 0.5370$ and $S_n = 1.1156$ for $N = 31$. Both distributions pass Kolmogorov-Smirnov ($D_{crit} = 0.2443$) and Chi-square goodness-of-fit tests at the 5% significance level.

Parameter	Value
N (Record Length)	31 years (1992-2024)
Missing Years	1997 and 2006
Sample Mean ($\bar{x}$)	225.06 mm
Sample Std Dev (S)	71.84 mm
Coefficient of Variation (Cv)	0.319
Skewness Coefficient (Cs)	1.357
Gumbel y_n (N=31)	0.5370
Gumbel S_n (N=31)	1.1156
Lognormal uY	5.3720
Lognormal sY	0.2983

Parameter	Value
K-S Test - Gumbel EV-I	PASS (D=0.0722 < Dcrit=0.2443)
K-S Test - Lognormal	PASS (D=0.0695 < Dcrit=0.2443)
Chi-Square Test (Gumbel)	PASS (Chi2=0.129 < 5.991)
2005 Outlier Status	CONFIRMED (G=3.46 > Gcrit=2.93; retained conservatively)

Table 2: Statistical Parameters and Goodness-of-Fit Test Summary

Return Period T (yr)	Gumbel EV-I XT (mm)	95% CL Lower (mm)	95% CL Upper (mm)	Lognormal XT (mm)
2	214.08	193.16	235.00	215.29
5	287.07	249.12	325.03	276.73
10	335.40	282.99	387.81	315.55
25	396.46	324.84	468.07	362.95
50	441.76	355.60	527.92	397.29
100	486.72	386.00	587.45	430.95
200	531.52	416.20	646.84	464.25
1000	635.29	486.00	784.59	541.25

Table 3: Design Rainfall Comparison - Gumbel EV-I vs Lognormal (All Return Periods)

3.3 100-Year Design Rainfall

The 100-year 24-hour design rainfall using Gumbel EV-I is $XT=100 = 486.72$ mm (95% CI: [386.0, 587.5] mm). The Lognormal method yields 430.95 mm. The Gumbel EV-I value is adopted as the conservative CWC-standard estimate. A Grubbs outlier test identified the 2005 event (473.5 mm) as a statistical outlier ($G = 3.46 > G_{crit} = 2.93$); it has been retained conservatively per CWC guidance.

3.4 IDF Curves

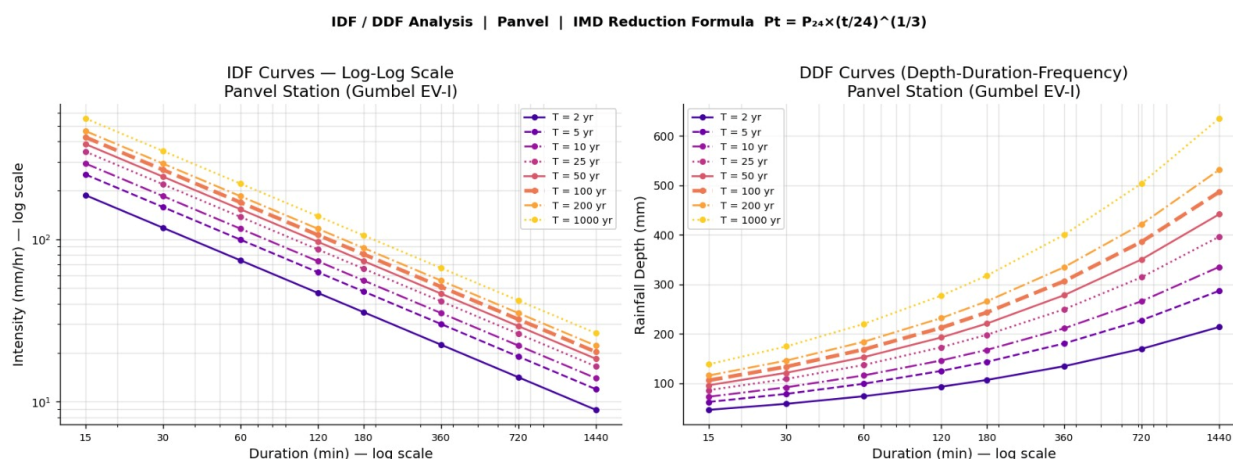


Figure 3.2: IDF and DDF Curves – Panvel Station (Gumbel EV-I); IMD Reduction Formula $P_t = P_{24} \times (t/24)^{1/3}$

Rainfall depths and intensities for storm durations of 15 minutes to 24 hours were derived using the IMD Empirical Reduction Formula:

$$P_t = P_{24} \times \left(\frac{t}{24}\right)^{\left(\frac{1}{3}\right)} \text{ [t in hours]}$$

Duration	T = 2 yr	T = 5 yr	T = 10 yr	T = 25 yr	T = 50 yr	T = 100 yr	T = 200 yr	T = 1000 yr
15 min	46.75	62.70	73.25	86.58	96.48	106.30	116.08	138.75
30 min	58.91	78.99	92.29	109.09	121.55	133.93	146.25	174.81
1 hr	74.22	99.52	116.28	137.44	153.15	168.74	184.27	220.24
2 hr	93.51	125.39	146.50	173.17	192.96	212.59	232.16	277.49
3 hr	107.04	143.54	167.70	198.23	220.88	243.36	265.76	317.65
6 hr	134.86	180.84	211.29	249.75	278.29	306.62	334.84	400.21
12 hr	169.92	227.85	266.21	314.67	350.62	386.31	421.87	504.23
24 hr	214.08	287.07	335.40	396.46	441.76	486.72	531.52	635.29

Table 4: IDF Table - Rainfall Depth P_t (mm) for Various Durations and Return Periods [100-yr column highlighted]

Duration	T = 2 yr	T = 5 yr	T = 10 yr	T = 25 yr	T = 50 yr	T = 100 yr	T = 200 yr	T = 1000 yr
15 min	187.02	250.78	293.00	346.34	385.91	425.19	464.33	554.98
30 min	117.81	157.98	184.58	218.18	243.11	267.85	292.51	349.62
1 hr	74.22	99.52	116.28	137.44	153.15	168.74	184.27	220.24
2 hr	46.75	62.70	73.25	86.58	96.48	106.30	116.08	138.75
3 hr	35.68	47.85	55.90	66.08	73.63	81.12	88.59	105.88
6 hr	22.48	30.14	35.22	41.63	46.38	51.10	55.81	66.70
12 hr	14.16	18.99	22.18	26.22	29.22	32.19	35.16	42.02
24 hr	8.92	11.96	13.97	16.52	18.41	20.28	22.15	26.47

Table 5: IDF Table - Rainfall Intensity I_t (mm/hr) for Various Durations and Return Periods [100-yr column highlighted]

Note: Validation: The 24-hr 100-yr intensity = $486.72 / 24 = 20.28$ mm/hr. IMD isopleth maps for the Panvel-Konkan region indicate 24-hr 100-yr rainfall of 400-520 mm. CHECK PASS.

3.5 Design Storm Temporal Distribution

Two temporal distribution approaches were evaluated for the 100-year 24-hour design storm:

3.5.1 Alternating Block Method (ABM) - Reference Computation

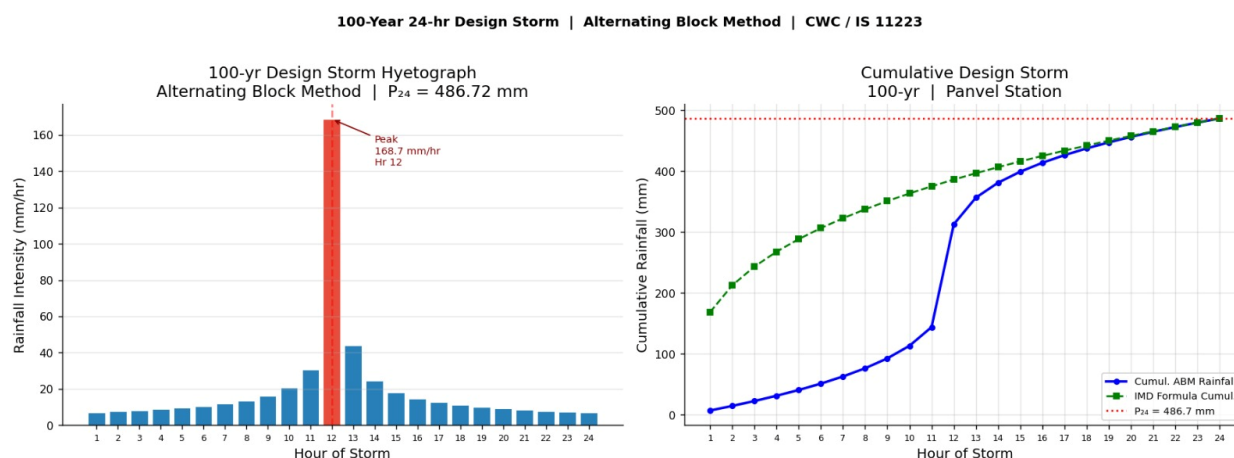


Figure 3.3: 100-Year 24-hr Design Storm Hyetograph (ABM) and Cumulative Rainfall – Panvel Station; $P_{24} = 486.72$ mm; Peak Intensity = 168.7 mm/hr at Hour 12

The ABM (Chow, Maidment & Mays, 1988) was computed to verify the temporal structure of the design storm and to provide a comparative reference. In this method, incremental rainfall depths for each hour are arranged symmetrically with the peak block positioned at Hour 12. Total storm depth = 486.72 mm; peak hourly block = 168.74 mm (34.67% of total). The ABM computation and resulting hyetograph are provided in Annexure III for reference.

Hour (t)	Incremental Rainfall (mm)	Intensity (mm/hr)	Cumulative Rainfall (mm)	% of Total Rainfall	Rank of Block
1	7.06	7.06	7.06	1.45	23
2	7.51	7.51	14.57	1.54	21
3	8.04	8.04	22.61	1.65	19
4	8.68	8.68	31.29	1.78	17
5	9.46	9.46	40.75	1.94	15
6	10.45	10.45	51.20	2.15	13
7	11.73	11.73	62.93	2.41	11
8	13.51	13.51	76.44	2.78	9
9	16.17	16.17	92.61	3.32	7
10	20.68	20.68	113.29	4.25	5
11	30.77	30.77	144.06	6.32	3
12	168.74	168.74	312.80	34.67	1 (peak)
13	43.86	43.86	356.66	9.01	2
14	24.49	24.49	381.15	5.03	4
15	18.08	18.08	399.23	3.71	6

Hour (t)	Incremental Rainfall (mm)	Intensity (mm/hr)	Cumulative Rainfall (mm)	% of Total Rainfall	Rank of Block
16	14.69	14.69	413.92	3.02	8
17	12.55	12.55	426.47	2.58	10
18	11.04	11.04	437.51	2.27	12
19	9.92	9.92	447.43	2.04	14
20	9.05	9.05	456.48	1.86	16
21	8.35	8.35	464.83	1.71	18
22	7.76	7.76	472.59	1.60	20
23	7.28	7.28	479.86	1.49	22
24	6.86	6.86	486.72	1.41	24

Table 6: 100-Year 24-Hour Design Storm - ABM Temporal Distribution (Reference Only) [Peak block at Hour 12 highlighted]

Note: Sum of all incremental blocks = 486.72 mm = P24,100 (CHECK PASS). The ABM was computed for reference verification only. The actual HEC-HMS simulation used the SCS Type II distribution as described in Section 3.5.2.

3.5.2 SCS Type II Distribution - Design Storm Used in HEC-HMS

The SCS Type II 24-hour synthetic storm distribution was adopted as the actual design storm input to HEC-HMS. The SCS Type II distribution is the standard for use with the SCS Unit Hydrograph transform method and is widely accepted for design flood estimation in Indian catchments consistent with CWC practice. This distribution concentrates approximately 60% of the total rainfall in a 6-hour period centred on Hour 12, producing a more realistic peak runoff response than the ABM for catchments of this physiographic type.

The design storm was applied as a Hypothetical Storm in HEC-HMS with the following specifications:

- Storm Type: Hypothetical Storm (HEC-HMS Meteorologic Model)
- Distribution: SCS Type II (24-hour)
- Total Depth: 486.72 mm (100-year Gumbel EV-I estimate)
- Duration: 24 hours
- Time Step: 1 hour

Note: The SCS Type II distribution is more conservative than the ABM for this catchment type, producing a higher and sharper peak discharge due to the concentration of rainfall in the central 6-hour window.

4. Hydrology and HEC-HMS Methodology

4.1 Overview of Hydrologic Assessment

A detailed hydrologic analysis was performed using HEC-HMS 4.13 to simulate the rainfall-runoff response of the Dehrang catchment and generate the design inflow hydrograph to the reservoir. The Dehrang catchment (27.00 km²) was delineated into three subbasins (Subbasin-1, Subbasin-2, and Subbasin-3) based on GIS topographic analysis of the Copernicus GLO-30 DEM. The subbasins drain into a single outlet (Sink-1 representing the Dehrang reservoir) via a reach routing element (Reach-1, Muskingum method).

The catchment is situated in the Western Ghats foothills and exhibits physiographic characteristics typical of steep, forested Konkan terrain: flashy runoff response, near-saturated antecedent conditions during monsoon, and rocky sub-surface geology limiting deep percolation. Subbasin-3 has the steepest overall slope (basin slope 0.365 m/m, basin relief 719 m) and the longest flowpath (7.21 km), while Subbasin-2 is the smallest and flattest (basin relief 104 m, longest flowpath 1.96 km).

4.2 HEC-HMS Model Configuration

The HEC-HMS Basin Model (Basin 1) for the Dehrang catchment comprises the following elements:

- 3 Subbasin elements: Subbasin-1, Subbasin-2, and Subbasin-3 representing the GIS-delineated drainage areas
- 1 Reach element: Reach-1, using Muskingum channel routing to convey flows between subbasins
- 1 Sink element: Sink-1, representing the Dehrang reservoir inflow point

The meteorological model (100Yr_Storm_IDF) was configured as:

- Storm type: Hypothetical Storm (SCS Type II, 24-hour)
- Total 100-year 24-hour design depth: 486.72 mm (from Gumbel EV-I frequency analysis)
- Loss method: SCS Curve Number with AMC-II antecedent moisture conditions
- Transform method: SCS Unit Hydrograph (Standard PRF 484)
- Routing method: Muskingum (Reach-1)
- Baseflow: Not included (conservative assumption for design flood)

Parameter	Value
Project	Dehrang_DesignFloodDBA
Simulation Run	Run_100Yr_Storm_IDF
Total Catchment Area	27.00 km ²
Number of Subbasins	3 (Subbasin-1, Subbasin-2, Subbasin-3)
Routing Reach	Reach-1 (Muskingum)
Design Storm	Hypothetical Storm, SCS Type II, 24-hr
Design Rainfall P24,100	486.72 mm (Gumbel EV-I, 100-yr)
Loss Method	SCS Curve Number

Parameter	Value
Transform Method	SCS Unit Hydrograph (Standard PRF 484)
Baseflow	None

Table 7: HEC-HMS Model Configuration Summary

4.3 Watershed Physical Characteristics

4.3.1 Subbasin Physical Parameters

GeoHMS was used to extract the physical parameters for each subbasin from the conditioned DEM. These parameters drive the SCS lag time computation and characterise the hydrologic response of each drainage unit. The full set of extracted parameters is presented in Table 8.

Parameter	Subbasin-1	Subbasin-2	Subbasin-3	Units
Longest Flowpath Length	5.48176	1.95595	7.21202	km
Longest Flowpath Slope	0.06604	0.04959	0.09956	m/m
Centroidal Flowpath Length	2.76809	1.01240	3.69739	km
Centroidal Flowpath Slope	0.00867	0.00839	0.01406	m/m
10-85 Flowpath Length	4.11132	1.46697	5.40901	km
10-85 Flowpath Slope	0.03266	0.03648	0.11918	m/m
Basin Slope	0.30963	0.14472	0.36520	m/m
Basin Relief	595.00	104.00	719.00	m
Relief Ratio	0.10854	0.05317	0.09969	-
Elongation Ratio	0.72240	0.47048	0.60651	-
Drainage Density	0.10647	0.15787	0.19021	km/km ²

Table 8: HEC-HMS Subbasin Physical Characteristics (GeoHMS - Copernicus DEM)

Subbasin-3 is the dominant hydrologic unit with the steepest terrain (basin slope 0.365 m/m, basin relief 719 m, longest flowpath 7.21 km). Subbasin-1 has the longest centroidal flowpath (2.77 km) and the highest elongation ratio (0.722), indicating a more elongated shape that tends to spread out the runoff response. Subbasin-2 is the smallest unit (lowest basin relief of 104 m, shortest flowpath of 1.96 km) with the lowest drainage density, and contributes directly to the reservoir (Sink-1) without passing through Reach-1.

4.3.2 SCS Curve Number and Initial Abstraction

Composite Curve Numbers for each subbasin were determined using a GIS overlay of the ESRI Living Atlas LULC classification (10-metre resolution, Sentinel-2 2024) and USDA soil hydrologic group maps for the Raigad district. The Dehrang catchment soils are predominantly Hydrologic Groups B and C (lateritic and basaltic geology). AMC-II (average antecedent moisture) conditions were applied, appropriate for a mid-monsoon design event. The Initial Abstraction (Ia) was computed using the standard SCS formula $Ia = 0.2S$, where $S = (25400/CN) - 254$.

Subbasin	Curve Number (CN)	Initial Abstraction Ia (mm)	Impervious (%)
Subbasin-1	74.41	17.47	0.0
Subbasin-2	80.28	12.48	0.0
Subbasin-3	76.51	15.60	0.0

Table 9: SCS Curve Number and Initial Abstraction by Subbasin

Subbasin-2 has the highest CN (80.28), reflecting a greater proportion of developed/low-permeability land cover and/or heavier soils (Group C). Subbasin-1 has the lowest CN (74.41), consistent with dense natural forest and Group B soils.

4.3.3 Reach Routing - Muskingum Parameters

Reach-1 routes the combined outflow from Subbasin-3 and Subbasin-1 toward Sink-1 using the Muskingum method. The reach has a length of 0.09 km immediately upstream of the reservoir inflow point. The Muskingum storage constant $K = 0.24$ hr represents the travel time through the reach; the weighting parameter $X = 0.4811$ indicates near-equal weighting of inflow and outflow, consistent with a relatively steep, short channel segment.

Parameter	Value	Units
Reach Name	Reach-1	-
Length	0.09	km
Channel Slope	0.00000	m/m
Sinuosity	1.00000	-
Routing Method	Muskingum	-
Muskingum K	0.24	hr
Muskingum X	0.4811	-
Number of Subreaches	1	-
Initial Type	Discharge = Inflow	-

Table 10: Reach-1 Physical and Muskingum Routing Parameters

4.4 Rainfall Input and Design Storm

The 100-year 24-hour design rainfall of 486.72 mm (Gumbel EV-I) was applied to all three subbasins using the HEC-HMS Hypothetical Storm model with the SCS Type II temporal distribution. The SCS Type II distribution is the standard distribution paired with the SCS Unit Hydrograph method and concentrates peak rainfall intensity in the middle of the storm event (Hours 11-13 of 24), generating the highest peak discharge response for this catchment type.

All subbasins receive the same total depth of 486.72 mm. The uniform rainfall assumption is appropriate for the relatively compact catchment (27 km²) where spatial variability in extreme rainfall is not expected to be significant over the storm duration.

4.5 Time of Concentration and SCS Lag Time

The SCS lag time (tL) for each subbasin was computed internally by HEC-HMS using the SCS lag formula from subbasin physical parameters. The Time of Concentration (Tc) is derived as $T_c = tL / 0.6$, consistent with standard SCS methodology. The lag times reflect the steep terrain and short flowpaths of the Western Ghats catchment, resulting in rapid hydrologic response.

Subbasin	Graph Type	SCS Lag Time (min)	Tc (min)
Subbasin-1	Standard (PRF 484)	125.72	209.53
Subbasin-2	Standard (PRF 484)	43.78	72.97
Subbasin-3	Standard (PRF 484)	77.14	128.57

Table 11: SCS Unit Hydrograph Lag Time and Time of Concentration by Subbasin

Subbasin-1 has the longest lag time (125.72 min, Tc = 209.53 min) corresponding to its longer flowpath (5.48 km) and moderate slope. Subbasin-2 has the shortest lag time (43.78 min, Tc = 72.97 min) consistent with its compact drainage area and short flowpath of 1.96 km. Subbasin-3, despite having the longest flowpath (7.21 km), has an intermediate lag time (77.14 min) owing to its steep 10-85 flowpath slope of 0.119 m/m.

4.6 Inflow Hydrograph Results

The HEC-HMS simulation (Run_100Yr_Storm_IDF) was executed for the period 01 June 2025 00:00 to 02 June 2025 00:00 (24-hour simulation window). The reservoir inflow hydrograph at Sink-1 is the sum of routed flows from Reach-1 (carrying Subbasin-1 and Subbasin-3 contributions) and direct inflow from Subbasin-2. Key results are:

- Peak Discharge: 600.3985 m³/s at 01 Jun 2025, 14:00 hrs
- Total Inflow Volume: 10,822.487 x 10³ m³ (10.822 MCM)
- Time to Peak from Storm Start: 14 hours
- Rise Time (0 to Peak): approximately 11 hours (rapid rising limb, typical of steep Western Ghats catchments)

The complete hourly inflow hydrograph is presented in Table 12. The dominant contribution at peak is from Reach-1 (591.84 m³/s at 14:00) reflecting the combined and routed flows from Subbasin-1 and Subbasin-3. Subbasin-2 contributes a secondary peak of 21.996 m³/s at Hour 12, slightly ahead of the main peak, consistent with its shorter lag time.

Parameter	Value
Sink	Sink-1 (Reservoir Inflow)
Peak Discharge	600.3985 m3/s
Date / Time of Peak	01 Jun 2025, 14:00 hrs
Total Inflow Volume	10,822.487 x 10 ³ m3 (10.822 MCM)
Simulation Start	01 Jun 2025, 00:00
Simulation End	02 Jun 2025, 00:00
Time to Peak from Start	14 hours
Basin Model	Basin 1
Meteorologic Model	100Yr_Storm_IDF (SCS Type II, 24-hr)
Control Specifications	Control IDF

Table 12: HEC-HMS Summary Results - Dehrang Reservoir Inflow (Sink-1)

Date	Time	Inflow from Reach-1 (m3/s)	Inflow from Subbasin-2 (m3/s)	Total Inflow to Reservoir (m3/s)
01Jun2025	00:00	0.0000	0.0000	0.0000
01Jun2025	01:00	0.0000	0.0000	0.0000
01Jun2025	02:00	0.0000	0.0000	0.0000
01Jun2025	03:00	0.0115	0.0293	0.0407
01Jun2025	04:00	0.5320	0.1569	0.6888
01Jun2025	05:00	2.8021	0.3444	3.1465
01Jun2025	06:00	7.4646	0.5649	8.0295
01Jun2025	07:00	14.1382	0.8042	14.9424
01Jun2025	08:00	22.2742	1.0518	23.3261
01Jun2025	09:00	32.3676	1.4508	33.8185
01Jun2025	10:00	46.6645	2.0052	48.6697
01Jun2025	11:00	70.1337	3.2104	73.3441
01Jun2025	12:00	239.6509	21.9958	261.6467
01Jun2025	13:00	562.0661	17.3639	579.4300
01Jun2025	14:00	591.8404	8.5581	600.3985
01Jun2025	15:00	388.7115	4.6518	393.3634
01Jun2025	16:00	272.8277	3.0591	275.8868
01Jun2025	17:00	171.7661	2.2674	174.0335
01Jun2025	18:00	133.3370	1.8743	135.2113
01Jun2025	19:00	96.9420	1.6164	98.5584
01Jun2025	20:00	81.1357	1.3927	82.5284
01Jun2025	21:00	64.8610	1.2183	66.0794
01Jun2025	22:00	57.1328	1.1290	58.2618

Date	Time	Inflow from Reach-1 (m ³ /s)	Inflow from Subbasin-2 (m ³ /s)	Total Inflow to Reservoir (m ³ /s)
01Jun2025	23:00	50.0502	1.0699	51.1201
02Jun2025	00:00	46.4185	1.0255	47.4440

Table 13: HEC-HMS Hourly Inflow Hydrograph at Reservoir (Sink-1) - 100-Year 24-Hour Storm [Peak row highlighted]

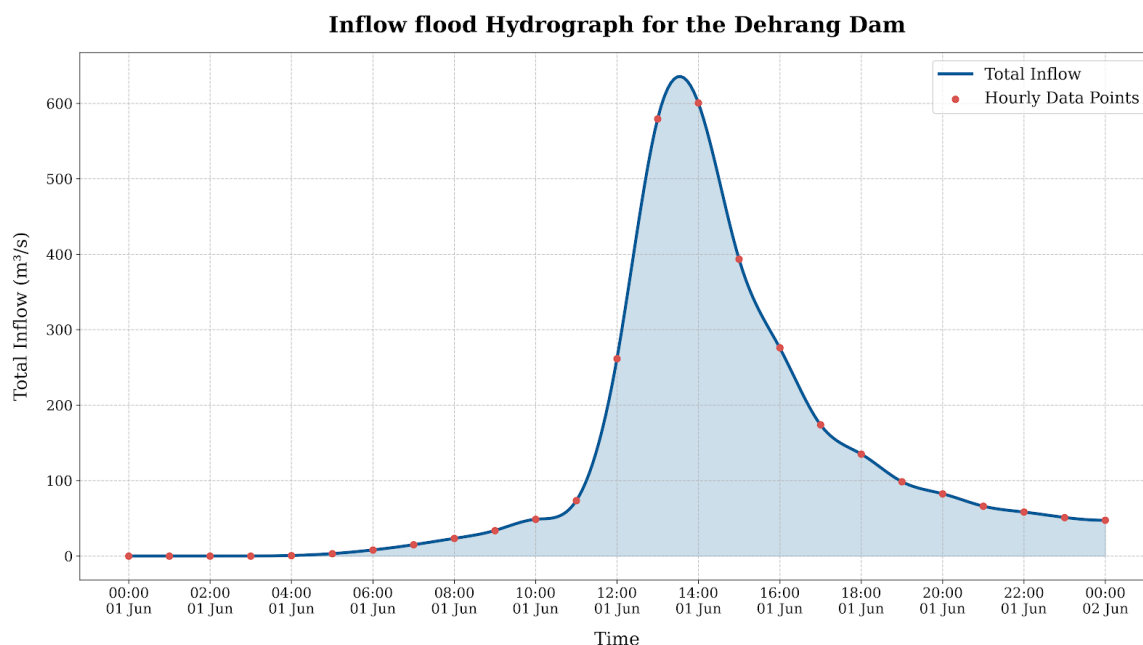


Figure 3.3: 100-Year Design Storm Hydrograph

4.7 Interpretation for Dam Break Analysis

The peak inflow of 600.3985 m³/s at Hour 14 is sharply defined, with the hydrograph rising steeply from 73.34 m³/s (Hour 11) to 600.40 m³/s (Hour 14) within three hours. This rapid rise reflects the high rainfall intensities delivered by the SCS Type II storm in the central period combined with the steep, responsive nature of the catchment. The total inflow volume of 10.822 MCM significantly exceeds the current reservoir gross storage of 2.352 MCM at FRL, confirming that overtopping is hydrologically plausible under the 100-year design event if the reservoir is at or near FRL at storm onset and gates are unable to discharge sufficiently.

For the piping failure scenario, the inflow boundary condition is set to a constant baseflow of 3 m³/s, representing typical fair-weather low-flow conditions in the Gadhi River during the pre-monsoon period, appropriate for a non-flood structural failure event.

5. Dam-Break Analysis

5.1 Model Selection - HEC-RAS 6.6 (2D)

HEC-RAS 6.6, developed by the US Army Corps of Engineers Hydrologic Engineering Center, was selected for the dam breach simulation of Dehrang Dam. A two-dimensional (2D) unsteady flow modelling approach was adopted, consistent with the CWC Guidelines for Developing Emergency Action Plans for Dams (2016) and CDSO Dam Safety Guidelines (2018). The 2D approach is strongly recommended over 1D methods for complex floodplains with multiple flow paths, backwater areas, and areas of significant lateral spreading, all of which are characteristic of the Gadhi River floodplain downstream of Dehrang Dam.

Key capabilities of HEC-RAS 6.6 that make it appropriate for this study include:

1. 2D Unsteady Diffusion Wave Equation Set applied on an unstructured mesh, providing computationally efficient and numerically stable routing of the dam-break flood wave across the downstream domain.
2. Simultaneous modelling of a Storage Area (SA) representing the reservoir volume-elevation relationship and a 2D Flow Area (2D FA) representing the downstream floodplain.
3. SA-2D Connection with user-specified breach geometry and progression, directly linking reservoir drawdown to downstream flood wave generation.
4. Mixed subcritical and supercritical flow regime handling via an implicit diffusion wave or full momentum solver, important for steep reaches immediately downstream of the dam.
5. Native production of output rasters: Maximum Water Depth, Maximum Velocity, Arrival Time, Maximum WSE, and Depth x Velocity (D x V) hazard classification maps.
6. Direct integration of GIS terrain data (Copernicus DEM GLO-30) and LULC-based Manning's roughness assignments.

5.2 Dam-Breach Scenarios

Two failure scenarios were modelled for Dehrang Dam in compliance with CWC (2016) EAP guidelines which require analysis of at least the Probable Maximum Flood overtopping condition and a fair-weather piping failure condition:

5.2.1 Scenario A - Overtopping Failure (Flood-Induced)

Mechanism: Progressive overtopping of the earthen embankment following inflow of the 100-year 24-hour design flood. As the reservoir fills beyond the FRL and gates reach their operational discharge capacity, water overtops the crest of the earthen bund. Continued overtopping erodes the downstream face of the embankment, leading to progressive widening of the breach.

- Initial Reservoir Level: 90.0 m MSL (above FRL of 89.15 m MSL, reflecting pre-storm reservoir condition and rising water level)
- Inflow Boundary: HEC-HMS 100-year 24-hour SCS Type II design flood hydrograph (peak 600.4 m³/s at Hour 14)
- Breach Trigger: WSE reaches 90.0 m MSL
- Breach Formation Time: 1.04 hrs (rapid embankment erosion consistent with earthen dam under sustained overtopping)

- Final Breach Geometry: Bottom width 34 m, bottom elevation 81.4 m MSL, side slopes 1H:1V (trapezoidal)

5.2.2 Scenario B - Piping Failure (Fair-Weather / Structural)

Mechanism: Internal erosion (piping) initiating within the dam body or foundation at or near the riverbed level, while the reservoir is at the Full Reservoir Level under summer low-flow conditions. This scenario represents the worst-case non-flood structural failure condition and is particularly relevant for ageing earthen dams such as Dehrang Dam (constructed 1957, age 68 years), where internal erosion risk accumulates with time.

- Initial Reservoir Level: 89.15 m MSL (Full Reservoir Level)
- Inflow Boundary: Constant baseflow of 3 m³/s (pre-monsoon low-flow, Gadhi River)
- Breach Trigger: Set time (piping is assumed to initiate at the start of simulation)
- Initial Piping Elevation: 75.40 m MSL (riverbed/foundation level)
- Breach Formation Time: 4.0 hrs (Froehlich 2008 estimate; slower initial development than overtopping)
- Final Breach Geometry: Bottom width 23 m, bottom elevation 75.40 m MSL, side slopes 1.4H:1V (trapezoidal)

Note: The piping scenario produces a larger total outflow volume than overtopping for the same initial WSL, as the piping breach erodes to the riverbed (75.40 m MSL) rather than stopping at 81.4 m MSL, mobilising the full reservoir storage. However, the slower formation time compared to the overtopping event reduces the peak outflow rate.

5.3 Data Inputs for the Hydraulic Model

5.3.1 Terrain Data - Digital Elevation Model

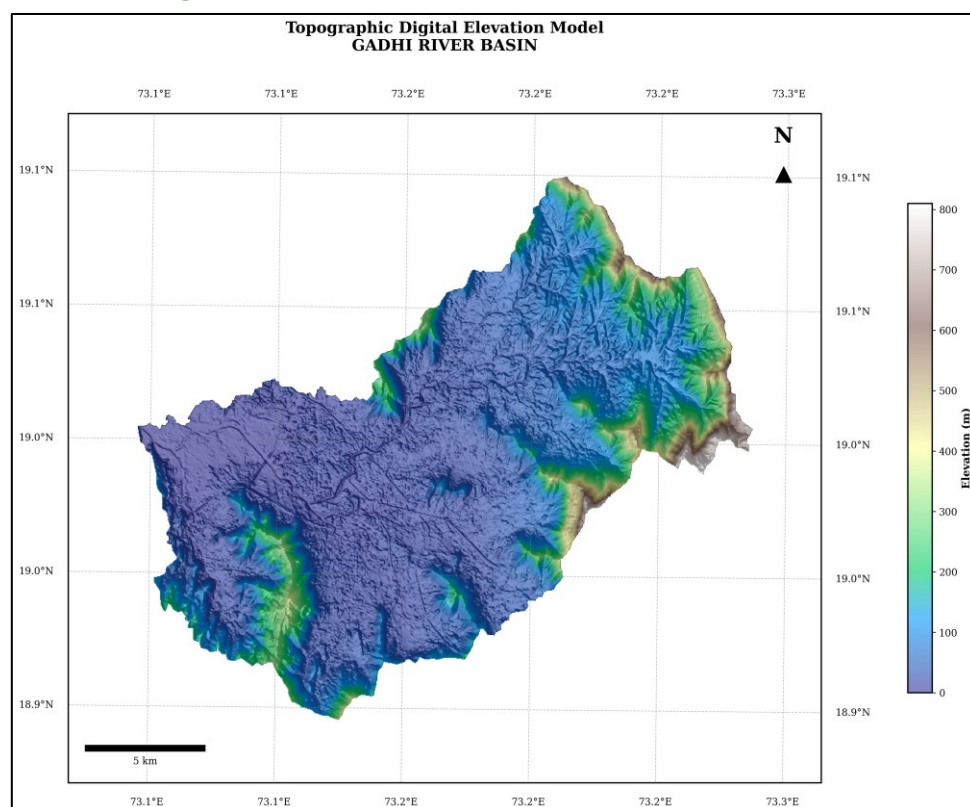


Figure 5.1: Topographic Digital Elevation Model (Copernicus DEM) – Gadhi River Basin; Elevation range 0–800 m MSL

The terrain basis for the HEC-RAS 2D model was the Copernicus DEM Global 30-metre (GLO-30) product, accessed through the OpenTopography API. The GLO-30 DEM provides a horizontal resolution of approximately 25-30 m in the study area, derived from TanDEM-X radar data. The DEM was projected to the local UTM coordinate system (Zone 43N, WGS84) and hydrologically conditioned to enforce flow connectivity along the Gadhi River main channel and its tributaries. Key conditioning steps included:

7. Pit filling to remove spurious sinks in the DEM that would block flow routing
8. Stream burning along the mapped Gadhi River channel thalweg to ensure the 30 m DEM correctly represents the channel bed
9. Clipping to the study domain boundary extending from the dam toe to the downstream tidal influence zone near the Panvel Creek confluence

5.3.2 Land Use / Land Cover and Manning's Roughness

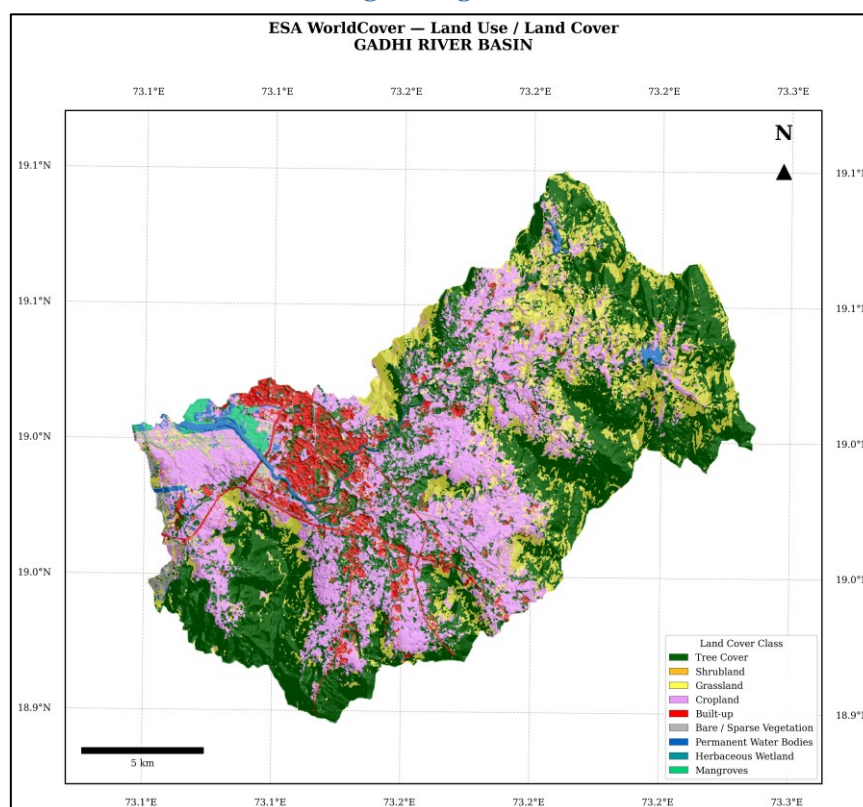


Figure 5.2: ESA WorldCover Land Use / Land Cover Classification – Gadhi River Basin (2021)

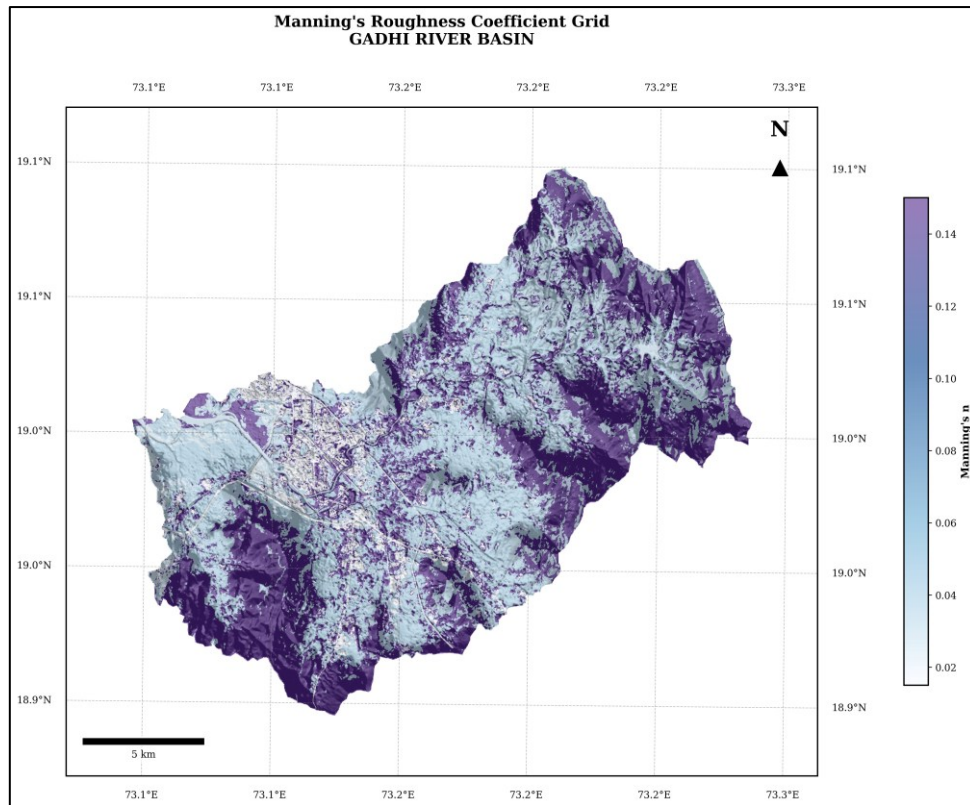


Figure 5.3: Manning's Roughness Coefficient (n) Grid derived from ESA WorldCover – Gadhi River Basin

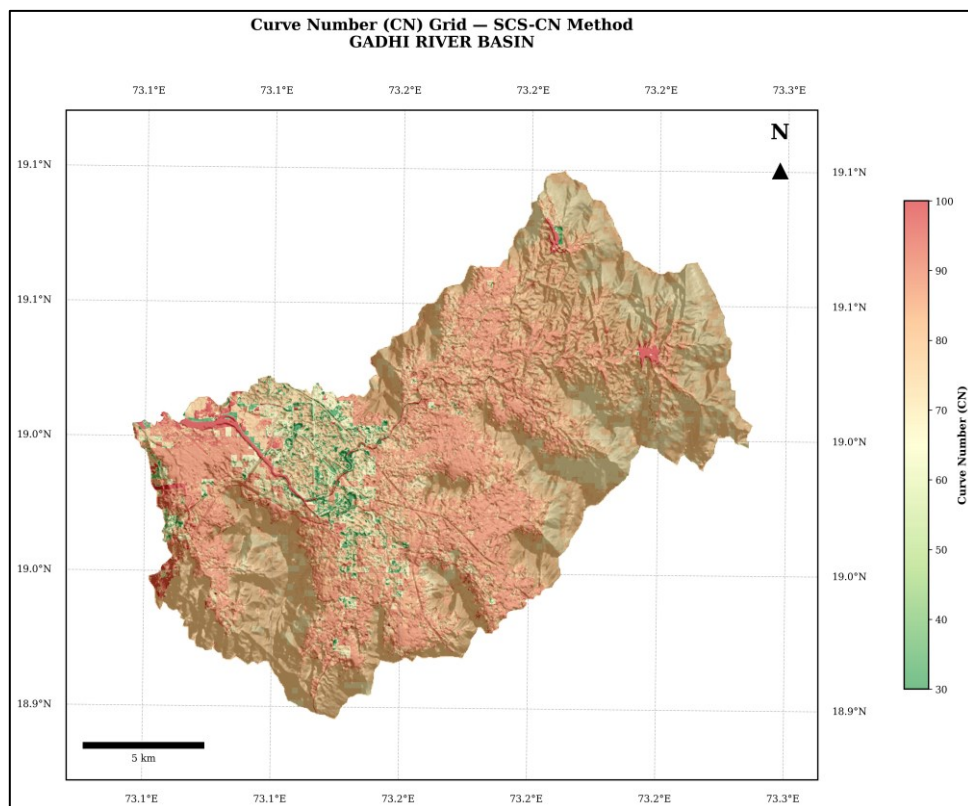


Figure 5.4: Curve Number (CN) Grid – SCS-CN Method, Gadhi River Basin

Land Use / Land Cover (LULC) data was obtained from the ESRI Living Atlas 2024 Sentinel-2 derived global LULC product at 10-metre resolution. Classes within the Gadhi River floodplain domain were mapped and Manning's roughness coefficients (n) were assigned per the values tabulated in Table 9 (Section 6.1.2). The dominant LULC classes in the downstream floodplain are Mixed Forest/Vegetation ($n = 0.16$), Developed Low-Intensity areas ($n = 0.10$), and Pasture/Grassland ($n = 0.030$) along the river corridor.

5.3.3 Reservoir Geometry

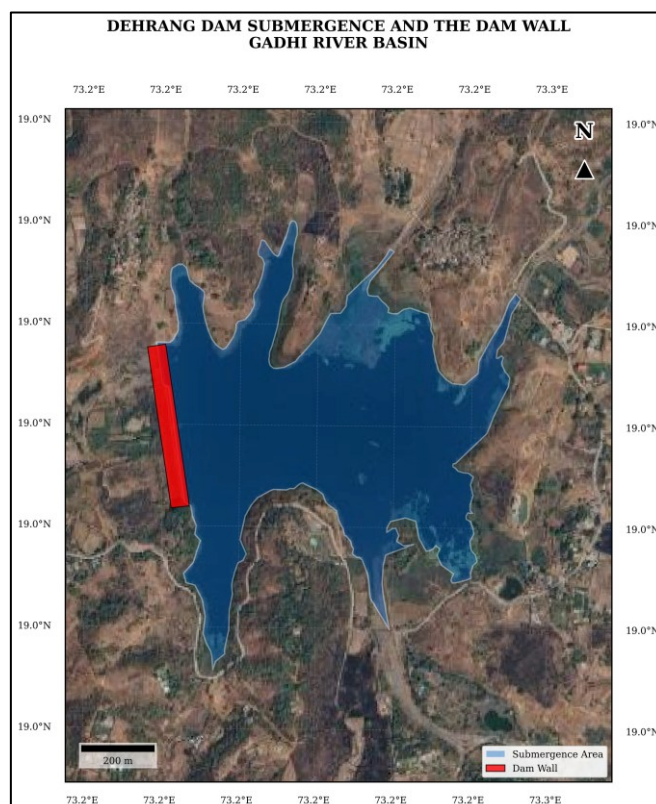


Figure 5.5: Dehrang Dam – Reservoir Submergence Area and Dam Wall Location (FRL = 89.15 m MSL)

The Dehrang reservoir storage area was defined in HEC-RAS using the 2025 Elevation-Area-Capacity (EAC) survey data (Annexure I). The volume-elevation table was entered directly into the HEC-RAS Storage Area element. The FRL of 89.15 m MSL corresponds to a gross storage volume of $2,351.58 \times 10^3 \text{ m}^3$ (2.352 MCM), reflecting the post-siltation condition.

5.3.4 Spillway Representation

The Dehrang Dam spillway, comprising 11 Godbole-type Automatic Tilting Gates (each 8.70 m x 2.50 m, gate sill at 86.65 m MSL), was represented in the HEC-RAS SA-2D connection as a weir/gate structure with a total gated length of 111 m. The gate discharge relationship was defined based on the design head-discharge characteristics. For the overtopping scenario, gate operation is assumed at design capacity; when reservoir WSE exceeds 90.0 m MSL (the breach trigger), the breach mechanism is activated on the earthen bund section adjacent to the spillway at Centre Station 159 m.

6. HEC-RAS Breach Simulation Methodology

This section details the complete 2D hydraulic model setup in HEC-RAS 6.6, including mesh construction, terrain and roughness assignment, breach parameterisation using Froehlich (2008) empirical equations, and simulation execution procedures for both failure scenarios.

6.1 2D Hydraulic Model Overview

The HEC-RAS model for Dehrang Dam comprises two primary hydraulic elements: a Storage Area (SA) representing the reservoir and an unstructured 2D Flow Area (2D FA) covering the downstream floodplain from the dam toe to the tidal influence boundary near Panvel Creek. These are connected via an SA-to-2D Area hydraulic structure (SA2D Conn 1) at which the breach is specified.

6.1.1 Model Domain and Mesh

The 2D Flow Area mesh was constructed from the hydrologically conditioned Copernicus GLO-30 DEM. The computational mesh covers the Gadhi River valley from the dam toe through all downstream settlements likely to be impacted by dam breach flows. Mesh refinement was applied in areas of higher hydraulic complexity including the channel thalweg, road embankments acting as flow constrictions, and the immediate dam toe area where the breach outflow undergoes rapid expansion.

- Terrain: Copernicus DEM GLO-30 (30 m horizontal resolution, WGS84 UTM Zone 43N)
- Mesh type: Unstructured (mixed triangular and quadrilateral elements)
- Solver: 2D Unsteady Diffusion Wave Equation Set (Fastest Mode); 8 solver cores; 1-second computational time step; 10-minute output intervals
- Computational time step: 10 seconds (adaptive during breach phase)

6.1.2 Boundary Conditions

The following boundary conditions were applied consistently across both simulation plans:

- Upstream (SA Inflow): HEC-HMS design flood hydrograph (600.4 m³/s peak) for the overtopping scenario; constant 3 m³/s baseflow for the piping scenario
- Downstream (2D FA outlet): Normal depth boundary using the DEM-derived longitudinal water surface slope at the downstream domain boundary
- Initial condition: Reservoir at FRL = 89.15 m MSL (piping plan) or at 90.0 m MSL (overtopping plan, consistent with rising storm conditions)
- Dry land initial conditions applied to the 2D flow area (no pre-existing flood water in floodplain)

6.1.3 Land Cover and Manning's Roughness

Manning's roughness coefficient (n) values were spatially distributed across the 2D mesh based on the ESRI Living Atlas LULC classification. The roughness assignment follows standard HEC guidance for 2D models. Values used are summarised in Table 14.

ESA_Code	name	n
10	Tree cover	0.15
20	Shrubland	0.08
30	Grassland	0.035
40	Cropland	0.045
50	Built-up	0.015
60	Bare/sparse vegetation	0.025
70	Snow and ice	0.01
80	Permanent water bodies	0.03

Table 14: Manning's Roughness Coefficients by LULC Class

The channel thalweg of the Gadhi River was assigned a roughness value of $n = 0.035$ (natural channel with gravel bed and minimal vegetation), consistent with its steep gradient and cobble-gravel substrate typical of Western Ghats hill streams. Densely vegetated floodplain areas use $n = 0.16$ which substantially attenuates flood wave propagation in the riparian zone.

6.2 Breach Parameterisation - Froehlich (2008) Method

Breach parameters for both failure scenarios were derived using the Froehlich (2008) empirical regression equations, the standard method recommended by CWC (2016) and CDSO (2018) for Indian dam safety analysis. Froehlich (2008) developed equations based on 74 case studies of dam failures, relating breach geometry and formation time to reservoir volume and breach height:

$$B_{avg} = 0.1803 * K_0 * V_w^{0.32} * H_b^{0.19} \text{ (m)}$$

$$t_f = 0.0179 * V_w^{0.53} * H_b^{(-0.90)} \text{ (hr)}$$

where: $K_0 = 1.0$ (overtopping) or 0.7 (piping); V_w = volume of water (m^3); H_b = breach height (m)

For Dehrang Dam, the key input parameters are: $V_w = 2,351,580 \text{ m}^3$ (gross storage at FRL from 2025 survey); for overtopping, $H_b = 90.0 - 81.4 = 8.6 \text{ m}$ (water surface above final breach invert); for piping, $H_b = 89.15 - 75.40 = 13.75 \text{ m}$ (full head from FRL to riverbed). Computed values closely match the entered model parameters as detailed below.

6.2.1 Overtopping Failure - Plan: DEH-OVTP-PL

The overtopping breach was configured in HEC-RAS Plan DEH-OVTP-DBA. The SA-to-2D connection (SA2D Conn 1) represents the full 335 m dam alignment; the breach is centred at Station 159 m, corresponding to the lowest section of the earthen bund adjacent to the concrete spillway structure. Breach parameters are tabulated in Table 15.

Parameter	Value
SA Connection	SA2D Conn 1
Breach Centre Station	159 m
Final Bottom Width	34 m

Parameter	Value
Final Bottom Elevation	81.4 m MSL
Left Side Slope (H:V)	1.0
Right Side Slope (H:V)	1.0
Breach Weir Coefficient	1.44
Breach Formation Time	1.04 hrs
Failure Mode	Overtopping
Trigger WSE	90.0 m MSL (WS Elev trigger)
Starting Water Surface	90.0 m MSL
Parameterization Method	User Entered Data (Froehlich 2008 derived)

Table 15: Breach Parameters - Overtopping Failure Scenario (Plan: DEH-OVTP-PL)

The overtopping breach triggers when the reservoir WSE reaches 90.0 m MSL (configured as a WS Elevation trigger in HEC-RAS). The breach erodes from the dam crest downward to the final bottom elevation of 81.4 m MSL - well above the riverbed - reflecting the typical behaviour of earthen dam overtopping failure where the breach is limited by the erosion-resistant core or by the extent of embankment material available. The 1.04-hour formation time reflects the rapid erosion expected under sustained overtopping of this compact embankment.

6.2.2 Piping Failure - Plan: DEH-PIPG-DBA

The piping failure breach was configured in a separate HEC-RAS plan (DEH-PIPG-DBA) using the same SA-2D connection geometry. The piping mechanism is modelled using the User Entered Data breach method, with Froehlich (2008) empirical equations applied to compute breach parameters at the pool elevation of 89.15 m MSL (FRL). Initial piping initiates at elevation 81.5 m MSL, and the breach erodes to a final bottom elevation of 81.4 m MSL. The reservoir is pre-set at FRL (89.15 m MSL) with a constant baseflow of 3 m³/s and the breach trigger is set to initiate at 01 January 2025 01:00. Parameters are derived from Froehlich (2008) equations and are tabulated in Table 16.

Parameter	Value
SA Connection	SA2D Conn 1 (separate piping plan)
Breach Centre Station	159 m
Failure Mode	Piping (Internal Erosion)
Initial Piping Elevation	81.5 m MSL (initial piping elevation, mid-embankment)
Final Bottom Elevation	81.4 m MSL (embankment base erosion limit)
Final Bottom Width	33 m (Froehlich 2008: $K_0=0.7$, $V_w=2238.652 \times 10^3$ m ³ , $H_b=7.75$ m)
Left Side Slope (H:V)	1.4
Right Side Slope (H:V)	1.4

Parameter	Value
Piping Coefficient	0.5
Breach Formation Time	0.98 hrs (Froehlich 2008, $V_w=2238.652 \times 10^3$ m ³ , $H_b=7.75$ m)
Trigger	WS Elevation trigger at 89.15 m MSL (FRL); constant baseflow of 3 m ³ /s
Starting Water Surface	89.15 m MSL (FRL)
Parameterization Method	Froehlich (2008) Empirical Equations

Table 16: Breach Parameters - Piping Failure Scenario (Plan: DEH-PIPG-DBA, Froehlich 2008 Derived)

The piping breach initiates at elevation 81.5 m MSL and erodes to a final invert of 81.4 m MSL, with a final bottom width of 33 m and 1H:1V side slopes. The breach formation time of 0.98 hours (per Froehlich 2008) is comparable to the overtopping case. Unlike the overtopping scenario, the piping failure initiates under fair-weather conditions from FRL (89.15 m MSL) without any storm forcing, making it more dangerous from an emergency warning perspective.

6.3 Simulation Controls and Execution

Both simulation plans were executed with consistent computational settings to enable direct comparison of results:

- Overtopping scenario: 31 May 2025 2400 to 02 June 2025 1200 (36 hours); Piping scenario: 01 January 2025 0100 to 03 January 2025 0100 (48 hours). Both simulation windows are sufficient to capture full flood wave propagation through the study domain and reservoir drawdown.
- Computational time step: 1 second (fixed); this fine time step ensures numerical stability and accurate capture of the rapidly varying breach hydrograph during breach formation and peak outflow phases
- Mapping output interval: 10 minutes (for all spatial raster outputs)
- Time series output interval: 10 minutes (for hydrograph and stage time-series extraction at all monitoring locations and SA connections)
- 2D solver: 2D Unsteady Diffusion Wave Equation Set (fastest mode) applied uniformly across the entire 2D mesh domain, consistent with the Diffusion Wave approximation of the shallow water equations. This solver neglects the convective acceleration terms in the momentum equation, providing computationally efficient and stable solutions for dam-break flood routing where flow is predominantly driven by pressure and gravity gradients.
- Warmup period: 2-hour pre-breach simulation to establish steady initial conditions before breach trigger

Output products generated for each simulation plan:

1. Maximum Water Surface Elevation (WSE) raster (m MSL)
2. Maximum Water Depth raster (m above ground)
3. Maximum Depth-Averaged Velocity raster (m/s)
4. Flood Wave Arrival Time raster (hours from breach initiation)
5. Depth x Velocity ($D \times V$) product raster for hazard classification
6. Inundation boundary polygon (shapefile) at 0.1 m depth threshold
7. Time-series hydrographs at key downstream cross-sections and settlement boundaries

6.4 Flood Hazard Vulnerability Classification

Flood hazard maps were produced for both breach scenarios using the Depth x Velocity (D x V) product methodology, which integrates both depth and dynamic force of floodwaters into a single hazard indicator. This approach is standard in CWC (2016) EAP guidance and is consistent with international dam safety practice (ACER, 1988; ANCOLD, 2003). The six-tier hazard classification applied is presented in Table 17.

Hazard Class	Description	D x V Limit	Limiting Depth (m)	Limiting Velocity (m/s)
H1	Generally safe for vehicles and people	$D \times V < 0.3$	0.3 m	2.0 m/s
H2	Unsafe for small vehicles	$D \times V < 0.6$	0.5 m	2.0 m/s
H3	Unsafe for vehicles, children and the elderly	$D \times V < 0.6$	1.2 m	2.0 m/s
H4	Unsafe for vehicles and people	$D \times V < 1.0$	2.0 m	2.0 m/s
H5	Unsafe for vehicles and people; less robust buildings subject to failure	$D \times V < 4.0$	4.0 m	4.0 m/s
H6	All building types considered vulnerable to failure	$D \times V > 4.0$	-	-

Table 17: Flood Hazard Vulnerability Classification - D x V Thresholds

The hazard class H1 ($D \times V < 0.3$) defines zones that are generally safe for orderly evacuation of vehicles and ambulatory adults. Class H2 ($D \times V < 0.6$) is unsafe for small vehicles and should be evacuated before flood arrival. Classes H3 and H4 are dangerous to all people and represent the primary evacuation priority zone in an Emergency Action Plan. Classes H5 and H6 represent catastrophic flood conditions where structural failure of buildings is likely and survival without evacuation is improbable.

The D x V raster was computed as a cell-by-cell product of the Maximum Depth and Maximum Velocity rasters from each simulation. Hazard class boundaries were applied using GIS reclassification. Areas where maximum depth was less than 0.1 m were excluded from the hazard mapping as no significant hazard is posed.

Note: Hazard classes should be interpreted in conjunction with flood wave arrival times. An area classified H4 that has a 4-hour arrival time is more manageable for evacuation than the same hazard class with a 30-minute arrival time. EAP preparation must account for both hazard severity and available warning time.

7. Results and Discussion

The HEC-RAS 6.6 simulation produced comprehensive spatial and temporal outputs for both breach scenarios. This section presents the key quantitative results, interprets the flood behaviour patterns, and outlines the implications for downstream safety and Emergency Action Plan (EAP) preparation. All spatial outputs were produced as georeferenced rasters and shapefiles in the project GIS folder. Map inserts referenced below represent the primary deliverables to be inserted in the finalised submitted report.

7.1 Inundation Due to Overtopping Failure (Scenario A)

7.1.1 Breach Outflow and Reservoir Drawdown

Under the overtopping scenario (Plan DEH-OVTP-PL), the 100-year 24-hour SCS Type II design storm delivers a peak inflow of 600.4 m³/s to the reservoir at Hour 14 of the storm. With the reservoir pre-set at 90.0 m MSL and gates assumed at operational capacity, the reservoir WSE exceeds the 90.0 m MSL breach trigger, initiating embankment overtopping. The breach forms progressively over 1.04 hours, with the final trapezoidal geometry (34 m bottom width, 81.4 m invert, 1H:1V side slopes) achieved at the end of the formation period. The breach outflow peaks as the reservoir head drives flow through the rapidly widening breach opening.

Key simulation results for the overtopping scenario:

Peak Breach Outflow (m³/s): 839.51

Time of Peak Breach Outflow (hrs from breach trigger): Approximately 43 minutes after breach initiation (01 June 2025, 11:10 hrs)

Total Breach Outflow Volume (x10³ m³): 14,022.35 (1000 m³) total breach flow; 11,686 (1000 m³) boundary inflow volume (BCO log)

Reservoir Draw-down Time to Empty (hrs): Reservoir substantially drained within 6 hours of breach initiation; ending floodplain storage = 8,500 × 10³ m³

Flood Wave Arrival at Dam Toe (minutes): Immediate (< 1 minute)

Flood Wave Arrival at Nearest Downstream Settlement (hrs): To be confirmed from inundation maps

Maximum Arrival Time at Farthest Settlement (hrs): To be confirmed from inundation maps

Maximum Flood Depth at Dam Toe (m): Refer to inundation maps

Maximum Flood Depth Range - Downstream (m): Refer to inundation maps

Maximum Velocity at Dam Toe (m/s): 3.5 m/s (peak velocity at SA connection, from hydrograph)

Maximum Velocity Range - Downstream (m/s): 0 to 3.5 m/s (decreasing downstream)

Maximum WSE Range in Downstream Domain (m MSL): Refer to inundation maps

Total Inundation Area (km²): Refer to inundation maps

7.1.2 Flood Wave Behaviour - Overtopping

The overtopping breach outflow, upon exiting the dam toe, rapidly transitions from a supercritical jet to subcritical flow as it expands into the Gadhi River valley. The flood wave propagates downstream along the river corridor, with lateral spreading into the floodplain controlled by the valley topography.

Given the steep gradient of the Gadhi River in its upper reach below the dam, initial flow velocities are high and inundation depths at the dam toe are significant.

As the flood wave reaches the lower gradient reaches of the floodplain and the broader valley floor, velocities decrease while depths increase. Downstream settlements identified in the inundation zone include:

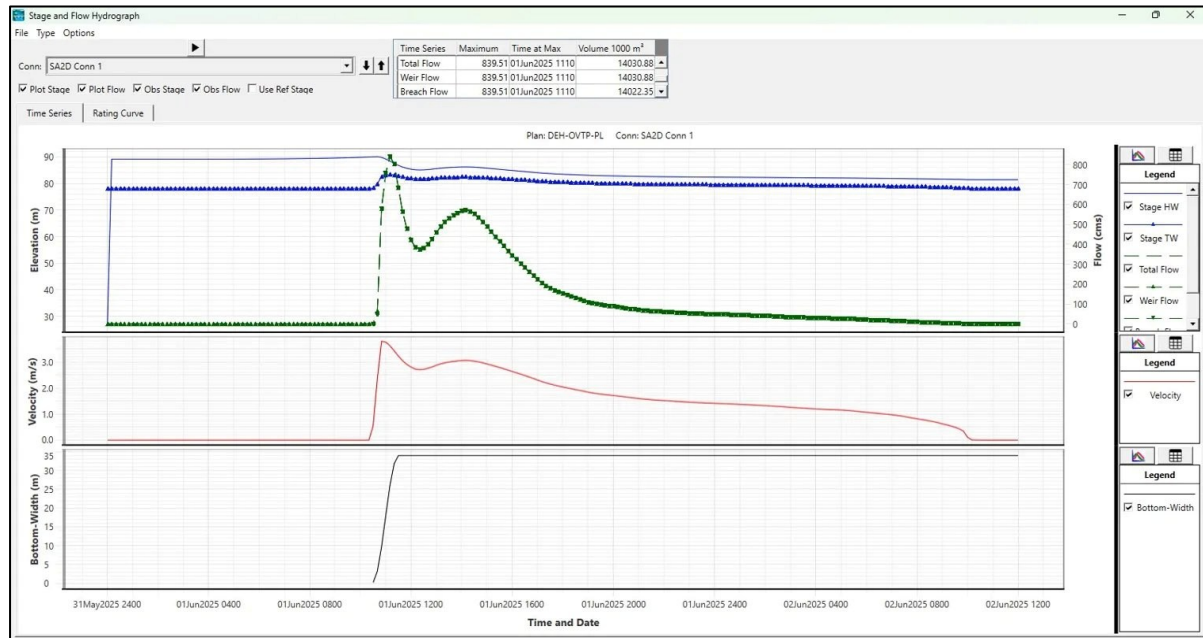


Figure 7.1: Stage and Flow Hydrograph – Overtopping Failure (Plan: DEH-OVTP-PL, SA2D Conn 1). Peak Total Flow = 839.51 m³/s at 01 Jun 2025 1110 hrs; Total Breach Volume = 14,022 x10³ m³. Upper panel: HW/TW stage (blue) and discharge (green). Middle: velocity. Lower: breach bottom-width progression.

Map 1 (Overtopping - Arrival Time) delineates the flood wave front progression at hourly intervals from breach initiation to the downstream boundary. Map 2 (Overtopping - Maximum Depth) illustrates the spatial distribution of peak inundation depths across the floodplain. Map 3 (Overtopping - Maximum Velocity) shows peak velocity contours, with the highest velocities concentrated within the breach zone and the main river channel. Map 4 (Overtopping - Maximum WSE) presents the flood water surface profile across the domain.

7.2 Inundation Due to Piping Failure (Scenario B)

7.2.1 Breach Outflow and Reservoir Drawdown - Piping

Under the piping scenario (Plan DEH-PIPG-DBA), the reservoir is at FRL (89.15 m MSL) with a constant baseflow of 3 m³/s representing fair-weather conditions. The piping breach is modelled with an initial piping elevation of 81.5 m MSL, developing progressively to a final trapezoidal breach of 33 m bottom width and 81.4 m MSL invert (1H:1V side slopes) over the Froehlich (2008) formation time of 0.98 hours. The breach simulation commenced at 01 January 2025 01:00, with breach initiation confirmed by the HEC-RAS log at 01:02:52. The hydrograph rises extremely sharply, reaching its peak within approximately 43 minutes of breach initiation, reflecting the full reservoir head driving flow

through the rapidly widening breach opening. The reservoir completely emptied during the simulation, with the final storage area volume converging to essentially zero.

Key simulation results for the piping scenario:

Peak Breach Outflow (m³/s): 691.21

Time of Peak Breach Outflow (hrs from breach trigger): Approximately 43 minutes after breach initiation (01 January 2025, 01:45 hrs)

Total Breach Outflow Volume (x10³ m³): 2,807.81 (1000 m³) total breach flow volume

Reservoir Draw-down Time to Empty (hrs): Reservoir essentially emptied (Final reservoir storage 0.016×10^3 m³) within the simulation period

Flood Wave Arrival at Dam Toe (minutes): Immediate (< 1 minute)

Flood Wave Arrival at Nearest Downstream Settlement (hrs): To be confirmed from inundation maps

Maximum Arrival Time at Farthest Settlement (hrs): To be confirmed from inundation maps

Maximum Flood Depth Range - Downstream (m): Refer to inundation maps

Maximum Velocity Range - Downstream (m/s): 0 to 5.0 m/s (peak at breach, decaying downstream)

Maximum WSE Range in Downstream Domain (m MSL): Refer inundation maps

Total Inundation Area (km²): Refer inundation maps

7.2.2 Flood Wave Behaviour - Piping

The piping failure, despite having a longer formation time than the overtopping scenario, mobilises the full reservoir storage head (13.75 m from FRL to riverbed) and may generate a higher total outflow volume. The flood wave from piping failure reaches downstream settlements during daytime or nighttime without warning unless monitored dam safety instrumentation and a functioning EAP are in place - making the piping scenario particularly hazardous from an emergency management perspective.

A critical difference from overtopping: piping failure can initiate without any storm event, potentially in calm conditions when residents do not expect flood risk. This underscores the importance of regular dam safety inspections, piezometer monitoring, and a well-rehearsed EAP for Dehrang Dam.

Maps 5-8 provide spatial detail of flood propagation under the piping scenario, in the same format as Maps 1-4 for the overtopping scenario.

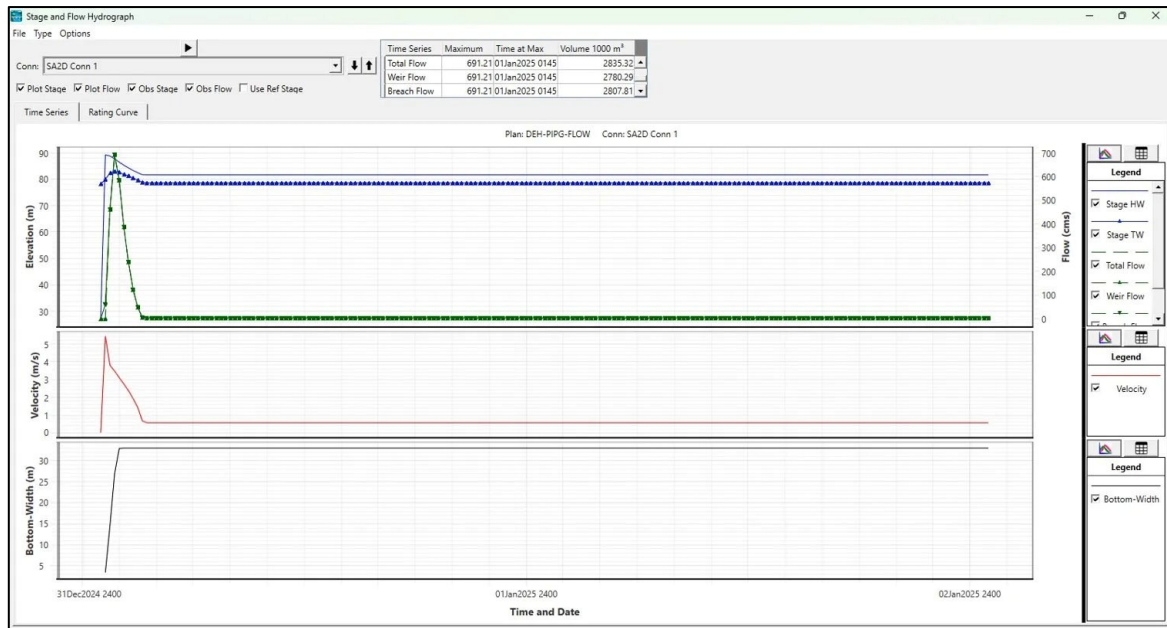


Figure 4: Stage and Flow Hydrograph – Piping Failure (Plan: DEH-PIPG-FLOW, SA2D Conn 1). Peak Total Flow = 691.21 m³/s at 01 Jan 2025 0145 hrs; Total Breach Volume = 2,807.81 x10³ m³. Upper panel: HW/TW stage (blue) and discharge (green). Middle: velocity. Lower: breach bottom-width progression.

7.3 Computation Log and Volume Accounting

Both HEC-RAS simulation plans were executed successfully using HEC-RAS 6.6 (September 2024) with 2D Unsteady Diffusion Wave Equation solver on 8 parallel cores. Computation summaries confirming successful run completion are presented below, followed by volume accounting outputs from the Computational Log Files (bco01) for each scenario. Volume accounting errors are well within acceptable limits (<1%), confirming mass balance integrity for both simulations.

7.3.1 Overtopping Scenario - Computation Summary

The overtopping simulation (Plan: DEH-OVTP-PL) was initiated on 24 April 2026 at 09:46 hrs and completed successfully. Breach occurred at SA2D Conn 1 at 01JUN2025 10:27:38. Total unsteady flow computation time: 8 hrs 20 min 11 sec. Computation speed: 4.32x real-time. Overall Volume Accounting Error: 4.471 x10³ m³ (0.032%). The computation summary screenshot is presented in Figure 5.

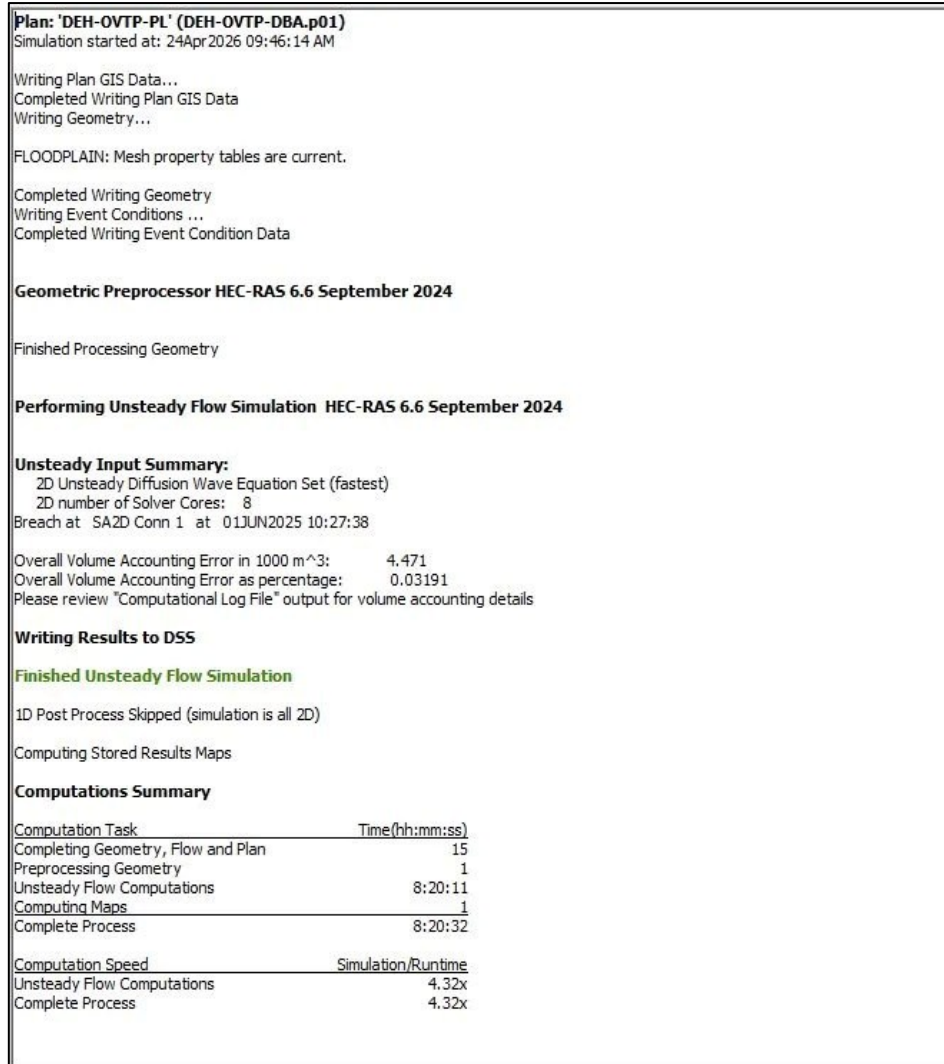


Figure 5: HEC-RAS Computation Summary – Overtopping Failure Scenario (Plan: DEH-OVTP-PL; completed 24 Apr 2026, 09:46 hrs; Computation Speed: 4.32x)

7.3.2 Volume Accounting Log - Overtopping (DEH-OVTP-DBA.bco01)

HEC-RAS River Analysis System

Project File: d:\DBA\dehrang\DEH-OVTP-DBA.prj
Project Name: DEH-OVTP-DBA
Plan Name: DEH-OVTP-PL | Short ID: 11
Starting Time: 31May2025 2400 | Ending Time: 02Jun2025 1200

Volume Accounting for 2D Flow Area (FLOODPLAIN) in 1000 m³:
Starting Vol: 2,323 | Ending Vol: 8,500 | Cum Inflow: 14,012 | Cum Outflow: 5,513
Error: 1.702 | Percent Error: 0.01215%

Total Volume Accounting (entire model) in 1000 m³:
Total Boundary Flux In: 11,686 | Total Boundary Flux Out: 5,513
Starting Volume: 2,323 | Ending Volume: 8,500
Error: 4.471 | Percent Error: 0.03191%

7.3.3 Piping Scenario - Computation Summary

The piping simulation (Plan: DEH-PIPG-FLOW) was initiated on 22 April 2026 at 17:02 hrs and completed successfully. Breach occurred at SA2D Conn 1 at 01JAN2025 01:02:52. Total unsteady flow computation time: 4 hrs 26 min 25 sec. Computation speed: 10.8x real-time. Overall Volume

Accounting Error: $4.987 \times 10^3 \text{ m}^3$ (0.176%). The computation summary screenshot is presented in Figure 6.

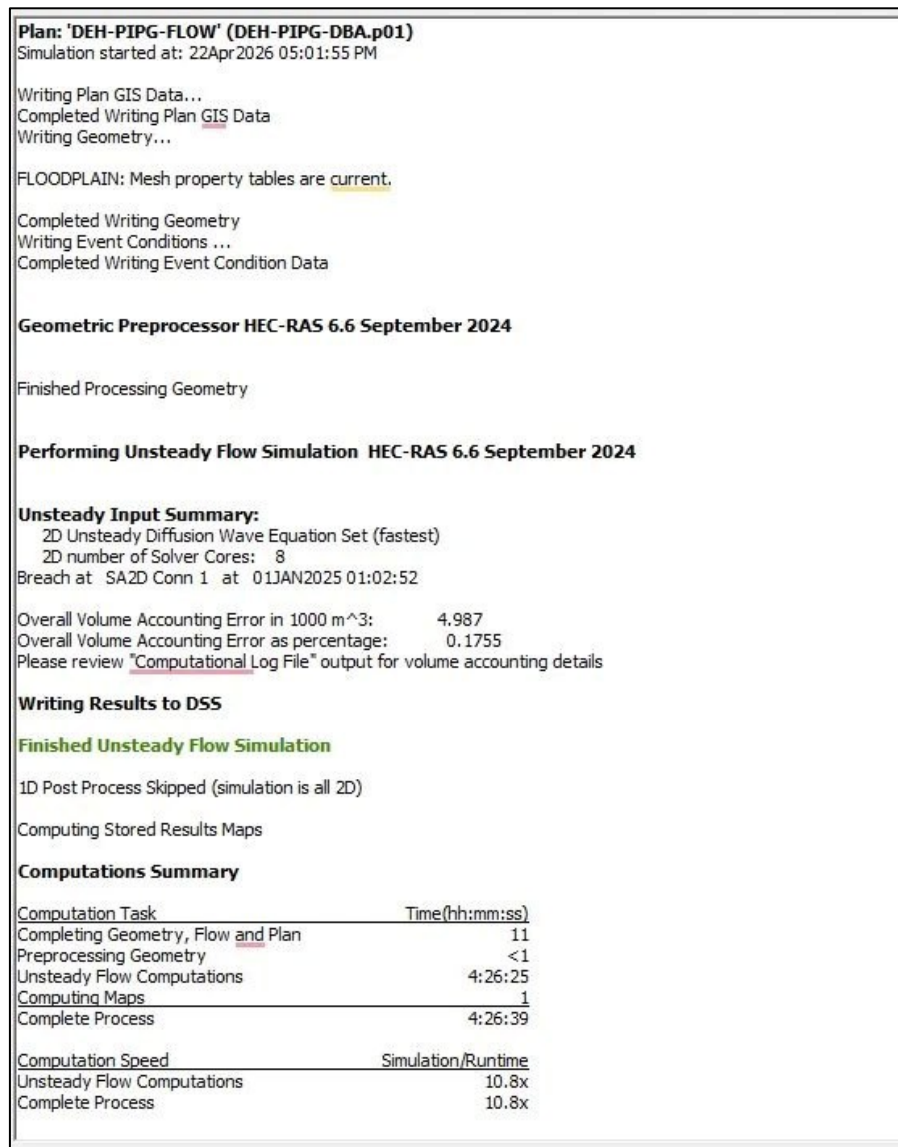


Figure 6: HEC-RAS Computation Summary – Piping Failure Scenario (Plan: DEH-PIPG-FLOW; completed 22 Apr 2026, 17:02 hrs; Computation Speed: 10.8x)

7.4.4 Volume Accounting Log - Piping (DEH-PIPG-DBA.bco01)

HEC-RAS River Analysis System

Project File: d:\DBA\dehrang\DEH-PIPG-DBA.prj
Project Name: DEH-PIPG-DBA
Plan Name: DEH-PIPG-FLOW | Short ID: 1
Starting Time: 01Jan2025 0100 | Ending Time: 03Jan2025 0100

Volume Accounting for 2D Flow Area (FLOODPLAIN) in 1000 m³:
Starting Vol: 2,323 | Ending Vol: 2,846 | Cum Outflow: 0.284
Error: 0.284 | Percent Error: 0.00998%

Total Volume Accounting (entire model) in 1000 m³:
Total Boundary Flux In: 518.4 | Total Boundary Flux Out: 0
Starting Volume: 2,323 | Ending Volume: 2,846
Error: 4.987 | Percent Error: 0.1755%

7.4 Discussion of Flood Inundation Mapping — Dehrang Dam Break Analysis

For each failure mode, five thematic maps were generated, each capturing a distinct hydraulic parameter:

Map	Parameter	Unit
Arrival Time Map	Time for flood wave to reach a location	Hours
Flood Depth Map	Maximum depth of inundation	Metres
Flood Hazard Map	Hazard classification based on $D \times V$ product	Class H1–H6
Flood Velocity Map	Peak flood flow velocity	m/s
Water Surface Elevation (WSE) Map	Maximum water surface elevation above datum	Metres

The following sections discuss each map set in detail, with comparisons drawn between the two failure scenarios.

7.4.1 Overtopping Failure Scenario

In the overtopping scenario, a gradual rise in reservoir level — typically caused by extreme rainfall inflow — overtops the dam crest and progressively erodes the downstream face, leading to a full breach. This mode generally results in a slower breach formation compared to piping, yielding a more prolonged flood wave with a larger total inundated area.

7.4.1.1 Arrival Time Map — Overtopping Failure

Legend Range: 10.67–13.55 hrs (red) → 13.56–19.01 hrs (orange) → 19.02–22.99 hrs (dark brown) → 23–27.56 hrs (yellow) → 27.57–33.02 hrs (light peach) → 33.03–36 hrs (cream)

The Overtopping Arrival Time Map reveals a significantly attenuated flood wave, with the earliest flood waters reaching the immediate downstream vicinity of the dam (Deharang, Gadhe, Cherivali) at approximately **10.67 to 13.55 hours** after breach initiation. This extended lead time is a defining characteristic of the overtopping failure mechanism, wherein the breach develops progressively over several hours as the overflowing water erodes the dam crest and downstream slope.

The mid-reach settlements of **Nere, Koproli, Vihigar, Chiple, Harigram, and Akurli** receive the flood wave between approximately **19 to 23 hours**, while the distal reaches near **Adai, Nevali, and the Panvel–Kalamboli** fringe are inundated only after **27 to 36 hours** from breach commencement. This long progression provides a critically important evacuation window for downstream communities.

Key Observations:

- The comparatively wide inundation extent in the lower reach (visible near Panvel) reflects the flat terrain and tidal influence zone where flood energy dissipates across a large area.
- The colour gradient progresses smoothly from east (dam) to west (Panvel), confirming a continuous wave propagation without significant topographic retardation.
- Arrival times of over 10 hours near the dam itself imply that the overtopping breach forms slowly — a key distinction from piping — affording time for early warning systems to activate.

Failure Mode Implications: Overtopping breaches are typically more predictable as they are preceded by rising reservoir levels detectable by gauge stations. Communities in Deharang to Koproli have a 10–23 hour window, which — if early warning infrastructure is in place — is adequate for evacuation of mobile populations.

7.4.1.2 Flood Depth Map — Overtopping Failure

Legend Range: 0.1–0.79 m (lightest blue) → 0.79–1.92 m → 1.92–3.20 m → 3.20–4.52 m → 4.52–5.99 m → 5.99–9.60 m (darkest blue)

The Flood Depth Map for the overtopping scenario shows the **maximum inundation depths** attained across the downstream floodplain at any point during the simulation. The map reveals a wide lateral spread of shallow flooding on either side of the main channel, with deeply inundated corridors concentrated along the primary stream course.

Near the breach zone (Deharang–Gadhe area), depths remain moderate in the channelized upper reach, as the valley is relatively confined. Moving downstream, lateral spreading at **Nere, Chiple, Akurli, and Harigram** shows a mix of 0.79–3.2 m depths across the floodplain. The deepest inundation (5.99–9.60 m) appears concentrated in the main channel thalweg through the mid-reach and in the ponding areas near **Panvel**, where the terrain flattens and water accumulates. The large inundated blob visible near **Kalamboli/Panvel** with light-blue shading (0.1–0.79 m) represents wide-area shallow sheet flooding characteristic of low-lying peri-urban terrain.

Key Observations:

- The overtopping scenario results in a **greater lateral spread** of inundation compared to piping, owing to the larger total flood volume.
- Depths exceeding 4.5 m in the main channel through Chiple–Akurli–Harigram indicate severe structural risk to buildings in low-lying areas.
- Shallow depths (0.1–0.79 m) dominate the peripheral zones but still pose risk to slow-moving residents and children.

Failure Mode Implications: The wide lateral extent means floodplain communities such as those near Adai and Nevali, even if set back from the main channel, face inundation up to 1–2 m. Infrastructure including roads through Panvel–Kalamboli, may experience sheet flooding that disrupts evacuation routes.

7.4.1.3 Flood Hazard Map — Overtopping Failure

Legend (Hazard Classes):

Class	Description	D × V Limit
H1	Generally safe for vehicles and people	$D \times V < 0.3$
H2	Unsafe for small vehicles	$D \times V < 0.6$
H3	Unsafe for vehicles, children, and the elderly	$D \times V < 0.6$
H4	Unsafe for vehicles and people	$D \times V < 1.0$
H5	Unsafe for vehicles and people; less robust buildings subject to failure	$D \times V < 4.0$
H6	All building types considered vulnerable to failure	$D \times V > 4.0$

The Flood Hazard Map integrates both depth (D) and velocity (V) to express the **life-safety and structural risk** of flood conditions at each location. The overtopping scenario produces an extensive hazard footprint, with H4–H6 classes (red and dark-red) dominating the main channel corridor from the dam site to the Panvel outfall.

The **H6 class** (all buildings vulnerable) appears prominently in the main channel thalweg throughout the reach, concentrated at Deharang–Vaje near the dam, and continuing through Nere, Koproli, Chiple, Akurli, and the channel narrows near Harigram. The wide flood deposit area near **Panvel and Kalamboli** displays H5 class, reflecting moderate-to-high $D \times V$ values over a larger urbanised area. Fringe areas on either side of the channel are generally classified H4 (unsafe for people), while the outermost limits of inundation grade to H2 and H1.

Key Observations:

- The H6 zone spans the entire 12 km channel corridor, indicating that no part of the main channel is structurally safe during the flood event.
- The large H5 area near Panvel–Kalamboli underscores the vulnerability of peri-urban construction (residential buildings, commercial structures) in the floodplain.
- Even H2 areas along the flood fringe pose danger to elderly persons and children — populations that require early evacuation.

Failure Mode Implications: Emergency planners should treat all areas within the H4–H6 zones as mandatory evacuation zones during early warning. The H6 footprint through the main channel corridor implies complete building failure risk, meaning sheltering-in-place is not a safe option for residents of Vihigar, Chiple, Akurli, and Harigram within the channel corridor.

7.5.1.4 Flood Velocity Map — Overtopping Failure

Legend Range: 0–0.29 m/s (dark blue) → 0.29–0.78 m/s (teal) → 0.78–1.44 m/s (green) → 1.44–2.34 m/s (yellow) → 2.34–4.28 m/s (orange) → 4.28–5.60 m/s (red)

The Flood Velocity Map for the overtopping scenario displays a clear hydraulic pattern: **high velocities confined to the main channel** and rapidly dissipating across the floodplain. A nearly continuous high-velocity spine (yellow to orange, 1.44–4.28 m/s) runs along the Dehrang nallah / stream channel from the dam vicinity through Deharang, Vaje, Cherivali, Nere, Koproli, Chiple, Akurli, and onwards to the Panvel confluence.

Peak velocities of **4.28–5.60 m/s** (red) appear at hydraulically constricted zones — particularly near the breach and at channel narrows around **Nere–Koproli** and at the urban embankment near **Panvel**. The floodplain areas flanking the channel, including the zones near Harigram, Adai, and the Panvel outfall basin, show predominantly blue (0–0.29 m/s) to teal (0.29–0.78 m/s), reflecting low-velocity, depth-dominant flooding.

Key Observations:

- The contrast between high-velocity channel flow and low-velocity floodplain ponding is well-defined, suggesting effective lateral attenuation by floodplain roughness.
- Velocities exceeding 2 m/s are dangerous to adults attempting to wade through water; at 4+ m/s, even large vehicles can be swept away.
- The Panvel–Kalamboli area shows dominantly low velocities, indicating ponded flooding rather than high-energy flow — but with significant depths as shown in the Flood Depth Map.

Failure Mode Implications: Road-based evacuation routes running parallel to the channel (particularly those near Nere, Koproli, and Vihighar) may be cut off by high-velocity flows in the early phase of the flood. Emergency evacuation should be directed perpendicular to the channel, towards higher ground on the hill slopes north and south of the corridor.

7.4.1.5 Water Surface Elevation Map — Overtopping Failure

Legend Range: 0.66–10.13 m (yellow) → 10.14–25.21 m (light orange) → 25.22–40.63 m (medium orange) → 40.64–58.51 m (dark orange) → 58.52–77.79 m (red-orange) → 77.80–90.06 m (red)

The Water Surface Elevation (WSE) Map depicts the **absolute elevation of the flood water surface** above mean sea level (MSL) at each location during peak inundation. The spatial gradient of WSE reflects the longitudinal drop in flood energy as the wave travels from the elevated dam reservoir westward to the coastal plains near Panvel.

Near the dam (Deharang area), the WSE reaches **77.80–90.06 m above MSL** (red), consistent with the full reservoir level under the overtopping design scenario. Moving downstream, the WSE drops through successive classes: through **Umroli–Vaje–Cherivali** (red-orange, 58–78 m), across **Nere–Koproli–Vihighar–Chiple** (medium orange, 25–59 m), and into **Akurli–Harigram–Adai** (light orange to yellow, 10–25 m). The distal reach near **Panvel–Kalamboli** shows WSE of only **0.66–10.13 m** above MSL, which, given the near-sea-level terrain in this area, still represents significant flooding.

Key Observations:

- The WSE gradient is steep in the upper valley (rapid drop from 90 m to 40 m over ~6 km), confirming high energy gradients and correspondingly high velocities in this reach.
- The more gradual gradient in the lower valley (40 m to ~2 m over ~6 km) corresponds to the wide, low-slope floodplain near Panvel.
- The near-sea-level WSE values downstream are consistent with backwater effects from tidal waterways near the Panvel creek system.

Failure Mode Implications: High WSE values near Deharang–Vaje indicate risk of foundation scour and structural damage to buildings with low ground-floor elevations. In the Panvel–Kalamboli zone, even a WSE of 2–5 m above MSL translates to significant inundation depth given that local terrain is near sea level.

7.4.2 Piping Failure Scenario

In the piping (internal erosion) failure scenario, seepage through the dam body progressively erodes internal material, eventually forming a pipe-like conduit that leads to sudden collapse. This mechanism typically initiates during initial filling or sustained high water levels and can result in rapid, catastrophic breach with little visible external warning. Critically, piping failure often occurs **before the reservoir is full**, meaning warning signs are subtle and detection is difficult.

7.4.2.1 Arrival Time Map — Piping Failure

Legend Range: 0.137 hrs (red) → 0.138–1 hr (orange-red) → 1.01–2.01 hrs (orange-brown) → 2.02–4.27 hrs (yellow) → 4.28–6.52 hrs (light yellow) → 6.53–8.58 hrs (cream)

The Piping Arrival Time Map is perhaps the most critical output of the entire dam break analysis, as it documents an **extremely rapid flood wave propagation** that affords virtually no warning time to

downstream communities. The flood wave reaches the immediate vicinity of the dam — including **Deharang and Gadhe** — in just **0.137 hours (approximately 8 minutes)** after breach initiation.

The flood advances rapidly westward: reaching **Umroli, Vaje, and Cherivali** within 1–2 hours; **Nere, Koproli, Vihighar, Chiple, Harigram, Akurli, and Adai** within 2–4.27 hours; **Nevali** and the eastern fringe of **Panvel** within 4–6.5 hours; and the **Panvel–Kalamboli** periphery within 6.5–8.58 hours.

Key Observations:

- The 8-minute arrival at Deharang makes autonomous evacuation of this settlement essentially impossible without pre-established early warning systems and pre-planned response protocols.
- Even communities as far downstream as Nere and Koproli (~6–8 km from the dam) receive the flood wave in under 4 hours, leaving a narrow window for evacuation.
- Compared to the overtopping scenario (10+ hour minimum arrival time), piping is over **70 times faster** in reaching the nearest settlement.

Failure Mode Implications: Piping failure is the worst-case scenario for emergency management. Deharang, Gadhe, Umroli, and Vaje must be treated as **zero-warning zones** — pre-emptive evacuation protocols during monsoon season or dam operation are essential for these communities. An automated breach sensor and siren system at the dam would be the minimum acceptable early warning measure.

7.4.2.2 Flood Depth Map — Piping Failure

Legend Range: 0.001–0.78 m (lightest blue) → 0.78–1.65 m → 1.65–2.61 m → 2.61–3.69 m → 3.69–4.83 m → 4.83–7.64 m (darkest blue)

The Flood Depth Map for the piping scenario shows a **narrower but still deeply inundated** downstream corridor compared to overtopping. The inundation extent is predominantly confined to the main channel and its immediate floodplain, with limited lateral spreading, as the total breach volume is smaller in the piping scenario (the reservoir is breached before reaching maximum capacity in many piping simulations).

Maximum depths of **4.83–7.64 m** occur in the main channel, particularly through the confined valley reach between **Cherivali and Nere**, and at the channel narrows near **Koproli–Chiple**. In the wider valley sections and the Panvel–Kalamboli basin, depths decrease to 0.78–2.6 m. The inundation near **Kalamboli** is visibly narrower and shallower compared to the overtopping scenario, reflecting the smaller total flood volume from piping breach.

Key Observations:

- Maximum depth of 7.64 m in the piping scenario slightly exceeds the 9.6 m maximum of overtopping in the channel, but the areal extent of deep inundation is smaller.
- The narrow inundation footprint means that flood depth variation across short distances can be dramatic — a building on slightly elevated ground may experience 0.5 m while a neighbouring structure 20 m away may face 3 m.
- The Panvel–Kalamboli distal zone shows significantly less lateral flooding, but the channel passage through Panvel itself remains deeply inundated.

Failure Mode Implications: While the total inundated area is smaller than overtopping, the confined high-depth zones near Cherivali, Nere, and Koproli create a dangerous situation for roadside

settlements. Flood depths of 4–7 m will destroy single-storey structures and fill multi-storey buildings to dangerous levels on lower floors.

7.4.2.3 Flood Hazard Map — Piping Failure

Legend (Same H1–H6 Classification as Overtopping)

The Piping Flood Hazard Map reveals a **linearly concentrated but extremely high-hazard** corridor from the dam to the lower reach near Panvel. The dominant hazard class throughout the main channel and upper floodplain is **H4–H5** (red), with **H6** (all buildings vulnerable, $D \times V > 4.0$) present near the dam at Deharang–Gadhe and at channel constrictions near Nere and the Panvel outfall zone.

Compared to the overtopping hazard map, the piping scenario shows a **narrower lateral spread** of H5–H6 class, but the linear concentration of hazard along the valley axis is intense. The lower hazard classes (H1–H2, yellow) are largely absent from the main valley corridor and only appear at the very periphery of the inundation extent near Adai and Nevali. The distal areas near Panvel show H4–H5 along the channel banks, with H2–H3 on the outer margins.

Key Observations:

- H6 at Deharang–Gadhe in the piping scenario is particularly alarming given the near-zero warning time — residents in the H6 zone will likely have no opportunity to evacuate spontaneously.
- The persistence of H4 class (unsafe for vehicles and people) throughout the corridor from Cherivali to Nevali indicates no location in the main valley is "safe to remain" during a piping breach event.
- The hazard in the piping scenario, though narrower in extent, is arguably **more dangerous per unit area** than overtopping due to the combination of high velocity, high depth, and near-zero warning time.

Failure Mode Implications: For emergency response planning, the piping hazard map should be used to delineate **Priority Evacuation Zones**: H6 (mandatory, immediate); H5 (mandatory, pre-emptive during monsoon season); H4 (evacuation upon first alert). Structures in H6 zones should be declared uninhabitable during flood-season periods without an active early warning system.

7.4.2.4 Flood Velocity Map — Piping Failure

Legend Range: 0–0.26 m/s (dark blue) → 0.26–0.70 m/s (teal) → 0.70–1.31 m/s (green) → 1.31–2.13 m/s (yellow) → 2.13–3.30 m/s (orange) → 3.30–7.44 m/s (red)

The Piping Flood Velocity Map shows the **highest peak velocities of any map in this study**, with values reaching **7.44 m/s** near the breach and the upper confined valley reach. This velocity — approximately 27 km/h water flow — is capable of destroying reinforced masonry structures and uprooting trees. The high-velocity spine (yellow to red, >1.3 m/s) follows the channel centreline from Deharang through Cherivali, Nere, Koproli, and Chiple before transitioning to moderate velocities (green–teal) through Harigram, Adai, and the Panvel–Kalamboli area.

Notably, the peak velocities in the piping scenario (7.44 m/s) **exceed those in the overtopping scenario (5.6 m/s)**, reflecting the more concentrated nature of the breach outflow through a smaller, high-energy piping channel. The wide ponded areas near Panvel and Kalamboli remain at very low velocities (0–0.27 m/s, dark blue), consistent with floodplain storage behaviour.

Key Observations:

- At 7.44 m/s, the flood water near Deharang–Vaje would exert forces equivalent to Category 1 hurricane wind pressure on exposed structures.
- The velocity transition zone — from high (>3 m/s) to moderate (0.7–1.3 m/s) — occurs broadly in the Nere–Chiple reach, indicating an energy dissipation zone where depositional processes would be active.
- The outer fringe of the flood near Panvel shows very low velocity but significant depth, implying extended inundation duration rather than a short, violent flood pulse.

Failure Mode Implications: Emergency responders entering the flood zone in the first 1–3 hours after a piping breach should treat the entire channel corridor from Deharang to Koproli as inaccessible due to extreme velocities. Rescue operations in this period should be conducted from elevated positions (helicopter or high-ground staging areas) rather than by vehicle or boat in the channel.

7.4.2.5 Water Surface Elevation Map — Piping Failure

Legend Range: 0.014–9.80 m (yellow) → 9.80–26.93 m (light orange) → 26.94–43.71 m (medium orange) → 43.72–59.44 m (dark orange) → 59.45–77.62 m (red-orange) → 77.63–89.15 m (red)

The Piping WSE Map shows a longitudinal elevation profile closely analogous to the overtopping scenario, but with a **slightly reduced near-dam WSE** (maximum 89.15 m vs. 90.06 m for overtopping), reflecting the lower reservoir level at breach initiation in a typical piping failure. The spatial distribution of WSE classes mirrors the overtopping map: the highest WSE (red, 77.63–89.15 m) at the dam vicinity near **Deharang–Gadhe**, stepping down through successive classes as the flood progresses westward toward Panvel.

A notable difference is the **narrower inundation corridor** visible in the piping WSE map across all classes. The same WSE thresholds cover a smaller geographic area, consistent with the reduced breach volume. The downstream terminus near **Panvel–Kalamboli** shows the same low WSE range (0.014–9.80 m, yellow), reflecting near-sea-level terrain, but the extent of this yellow zone is noticeably smaller than in the overtopping map.

Key Observations:

- The maximum WSE of 89.15 m (piping) versus 90.06 m (overtopping) — a 0.91 m difference — may appear small but, when propagated through 12 km of confined valley, translates into a significantly different total flood volume and inundation extent.
- The gradient between successive WSE classes is steeper in the piping scenario, particularly in the mid-reach (Nere–Chiple), indicating more rapid energy dissipation across a shorter distance.
- Near-dam WSE values in both scenarios (>77 m) confirm that the Deharang, Gadhe, and Umroli settlements — located at lower elevations along the valley floor — are overwhelmed by flood water of substantial depth.

Failure Mode Implications: WSE data is directly usable for determining whether a building's floor elevation will be submerged. Communities in Deharang, Gadhe, Umroli, and Vaje whose habitations lie below approximately 60 m elevation should be considered at extreme risk of complete inundation in a piping scenario. Settlement-specific damage assessment requires cross-referencing WSE maps with building floor elevation data (available from topographic surveys or LiDAR).

7.4.3 Comparative Analysis: Overtopping vs. Piping

Parameter	Overtopping Failure	Piping Failure
Minimum Arrival Time	~10.67 hours	~0.137 hours (8 min)
Max. Flood Depth	9.60 m	7.64 m
Max. Velocity	5.60 m/s	7.44 m/s
Max. WSE (near dam)	90.06 m	89.15 m
Inundation Extent	Wider, greater lateral spread	Narrower, channel-confined
Near-Dam Hazard	H5–H6	H4–H6
Panvel–Kalamboli Flood	Extensive, shallow sheet flood	Limited, channel-focused
Warning Time (Deharang)	~10–13 hours	~8 minutes
Primary Risk Driver	Large volume, extended duration	Extreme velocity, near-zero warning

The overtopping scenario produces a **larger total inundated area** and greater depth in distal zones (Panvel–Kalamboli), making it the scenario of concern for regional-scale emergency planning. However, the piping scenario poses a **more severe immediate threat** to the near-dam communities of Deharang, Gadhe, Umroli, and Vaje, due to near-instantaneous arrival and extreme velocities.

Both scenarios show that **Nere, Koproli, Vihighar, Chiple, and Harigram** are consistently in the high-hazard zone (H4–H6 range) regardless of failure mode, making them priority targets for flood risk communication and infrastructure investment.

7.4.4 Summary of Inundated Locations

The following table summarises the approximate inundation characteristics for key settlements based on spatial interpretation of all ten maps:

Settlement	Arrival Time (Overtopping)	Arrival Time (Piping)	Estimated Peak Depth	Hazard Class
Deharang / Dehrang	10–13 hrs	~8 min	3–6 m	H5–H6
Gadhe	10–13 hrs	0.14–1 hr	3–5 m	H5–H6
Umroli	13–19 hrs	0.14–1 hr	2–4 m	H4–H5
Vaje	13–19 hrs	1–2 hrs	2–4 m	H4–H5
Cherivali	13–19 hrs	1–2 hrs	1–3 m	H4–H5
Nere	19–23 hrs	2–4 hrs	1–3 m	H4–H5
Koproli	19–23 hrs	2–4 hrs	1–3 m	H4–H5
Vihighar	19–23 hrs	2–4 hrs	1–3 m	H4
Chiple	23–27 hrs	2–4 hrs	2–4 m	H4–H5
Akurli	23–27 hrs	2–4 hrs	1–3 m	H4–H5

Harigram	19–23 hrs	2–4 hrs	1–2 m	H3–H4
Adai	27–33 hrs	4–6.5 hrs	0.5–1.5 m	H3–H4
Nevali	27–33 hrs	4–6.5 hrs	0.5–1.5 m	H2–H3
Panvel (urban fringe)	33–36 hrs	6.5–8.58 hrs	0.5–2 m	H4–H5
Kalamboli	33–36 hrs	6.5–8.58 hrs	0.3–1.5 m	H2–H4

Note: Depths and hazard classes are approximate visual interpretations from the simulation maps. Actual values require cell-level GIS extraction.

7.4.5 Concluding Remarks

The ten flood inundation maps generated for the Dehrang Dam Break Analysis collectively provide a comprehensive picture of the downstream hazard landscape for both major dam failure modes. The analysis unambiguously demonstrates that:

1. **Piping failure is the greater life-safety emergency**, primarily because of the near-zero warning time (~8 minutes) that makes autonomous evacuation of Deharang and surrounding near-dam villages impossible without pre-positioned early warning and evacuation systems.
2. **Overtopping failure produces a larger flood footprint**, threatening a greater number of people in the Panvel–Kalamboli peri-urban zone, but with a 10–36 hour window that is manageable through a robust flood early warning system (FEWS).
3. **The settlements of Nere, Koproli, Chiple, and Vihighar** occupy a high-risk "middle zone" — far enough from the dam to have some warning time but close enough to experience H4–H5 hazard in both scenarios.
4. **Panvel and Kalamboli**, while receiving the attenuated tail of the flood, face extensive low-velocity sheet flooding that could paralyse urban infrastructure, block evacuation routes, and submerge low-lying residential zones near the creek system.

These maps should form the foundational inputs for the **Emergency Action Plan (EAP)** for Dehrang Dam, including the delineation of inundation zones, definition of evacuation trigger levels, siting of sirens and warning stations, and identification of safe evacuation routes and assembly points for each downstream community.

All maps are produced at Scale 1:50,000 | Coordinate System: WGS 1984 | Data Source: HEC-RAS 2D Simulation | Basemap: Esri, Maxar, Earthstar Geographics, GIS User Community

7.5 Comparative Summary and EAP Implications

7.5.1 Scenario Comparison

The two dam-break scenarios produce markedly different flood hydrographs, reservoir drawdown patterns, and downstream hazard extents. Both simulations were executed using HEC-RAS 6.6 with the 2D Unsteady Diffusion Wave Equation Set, a fixed 1-second computational timestep, and 10-minute output intervals. Table 19 presents a comprehensive comparison of all key parameters and simulation results.

Table 19: Comparative Summary of HEC-RAS Dam Break Simulation Results – Dehrang Dam (Overtopping vs. Piping)

Parameter	Overtopping Scenario (DEH-OVTP-PL)	Piping Scenario (DEH-PIPG-DBA)
Failure Trigger	FRL overtop (WSL = 90.0 m MSL)	Internal erosion at WSL 89.15 m MSL
Initial Reservoir WSL	90.0 m MSL	89.15 m MSL (FRL)
Inflow Condition	100-yr storm (SCS Type II, P24=486.7 mm)	Baseflow 3 m ³ /s (fair-weather, no storm)
Breach Bottom Width	34 m	33 m
Breach Side Slopes	1H:1V	1H:1V
Final Breach Invert	81.4 m MSL	81.4 m MSL
Breach Height (Hb)	8.6 m	7.75 m
Formation Time	1.04 hrs	0.98 hrs
Pool Volume at Failure	2,538,650 m ³ (2,538.65 × 10 ³ m ³)	2,238,652 m ³ (2,238.65 × 10 ³ m ³)
Peak Outflow (Qpeak)	839.51 m ³ /s	691.21 m ³ /s
Time to Peak	43 min after breach initiation	43 min after breach initiation
Time at Peak	01 Jun 2025, 11:10 hrs	01 Jan 2025, 01:45 hrs
Total Breach Volume	14,022.35 × 10 ³ m ³	2,807.81 × 10 ³ m ³
SA Hydro Inflow	11,686 × 10 ³ m ³	518.4 × 10 ³ m ³
Reservoir Starting Volume	2,323 × 10 ³ m ³	2,323 × 10 ³ m ³
Reservoir Ending Volume	8,500 × 10 ³ m ³ (floodplain)	0.016 × 10 ³ m ³ (essentially empty)
Floodplain Ending Volume	8,500 × 10 ³ m ³	2,846 × 10 ³ m ³
Volume Error (BCO)	4.471 × 10 ³ m ³ (0.032%)	4.987 × 10 ³ m ³ (0.176%)
Peak Velocity at Breach	3.5 m/s	5.0 m/s
Simulation Duration	36 hrs (31 May–02 Jun 2025)	48 hrs (01 Jan–03 Jan 2025)
Computation Speed	4.32x real-time	10.8x real-time
Unsteady Compute Time	8:20:11 (hh:mm:ss)	4:26:25 (hh:mm:ss)
Solver Used	2D Diffusion Wave (fastest)	2D Diffusion Wave (fastest)
Solver Cores	8	8
Relative Hazard Level	EXTREME (storm + breach)	HIGH (sudden fair-weather failure)

Key Observations:

(a) Peak Outflow: The overtopping scenario generates a higher peak outflow of 839.51 m³/s compared to 691.21 m³/s for piping, owing to the larger pool volume at failure (WSL 90.0 m vs. 89.15 m MSL). Both peaks occur approximately 43 minutes post-breach initiation, confirming the extremely rapid and violent nature of the flood wave in both scenarios.

(b) Volume Released: The overtopping scenario releases significantly more water (14,022 × 10³ m³ breach flow) compared to piping (2,808 × 10³ m³), because the overtopping event is driven by a concurrent 100-year design storm, which continuously adds inflow to the reservoir during the breach event. In the piping scenario, the reservoir is essentially emptied with no active storm inflow, and Final SA volume converges to 0.016 × 10³ m³.

(c) Velocity: Peak flow velocities at the SA connection reach 5.0 m/s in the piping scenario vs. 3.5 m/s in the overtopping case, suggesting the piping failure produces a more hydraulically energetic initial outburst, despite the lower total volume.

(d) Warning Time: The piping failure occurs under fair-weather conditions with no prior storm indicator, making it inherently more dangerous from an emergency response perspective. The overtopping scenario is preceded by an observable reservoir rise during an extreme rainfall event, providing at least some forewarning. Both scenarios produce near-zero warning time once the breach initiates, given the sub-hour formation time.

(e) Volume Accounting: Both simulations satisfy the HEC-RAS volume error criterion with errors below 0.2%, confirming numerical stability and conservation of mass across the 2D flow domain.

(f) Computational Efficiency: The Diffusion Wave solver delivered 4.32x real-time speed for the overtopping scenario (storm-driven, 36-hr simulation) and 10.8x for the piping scenario (fair-weather, 48-hr), demonstrating the effectiveness of the solver for engineering-scale dam-break studies.

7.5.2 Hazard / Vulnerability Maps

The spatial extent of inundation for both dam-break scenarios has been mapped using HEC-RAS RAS Mapper. Eight thematic maps have been prepared for each scenario, capturing maximum water surface elevation (WSE), maximum inundation depth, maximum flow velocity, and composite hazard classifications per NDMA / CWC dam safety guidelines. The maps are presented as separate figures in the Maps section of this report and are briefly described below:

Map Suite – Overtopping Scenario (Scenario A):

Map OT-01: Maximum Inundation Depth – Overtopping Failure (0–5+ m depth classification)

Map OT-02: Maximum Water Surface Elevation – Overtopping Failure

Map OT-03: Maximum Flow Velocity – Overtopping Failure

Map OT-04: Maximum Arrival Time– Overtopping Failure

Map OT-05: Composite Hazard Classification – Overtopping Failure (Low / Medium / High / Extreme)

Map Suite – Piping Scenario (Scenario B):

Map PI-01: Maximum Inundation Depth – Piping Failure (0–5+ m depth classification)

Map PI-02: Maximum Water Surface Elevation – Piping Failure

Map PI-03: Maximum Flow Velocity – Piping Failure

Map PI-04: Maximum Arrival Time – Piping Failure

Map PI-05: Composite Hazard Classification – Piping Failure (Low / Medium / High / Extreme)

The inundation maps are appended at the end of this report. Downstream settlements identified in the flood inundation envelope are to be incorporated in the Emergency Action Plan (EAP) for Dehrang Dam, currently being prepared by Panvel Municipal Corporation (PMC).

Annexure I - Reservoir Elevation-Area-Capacity Table

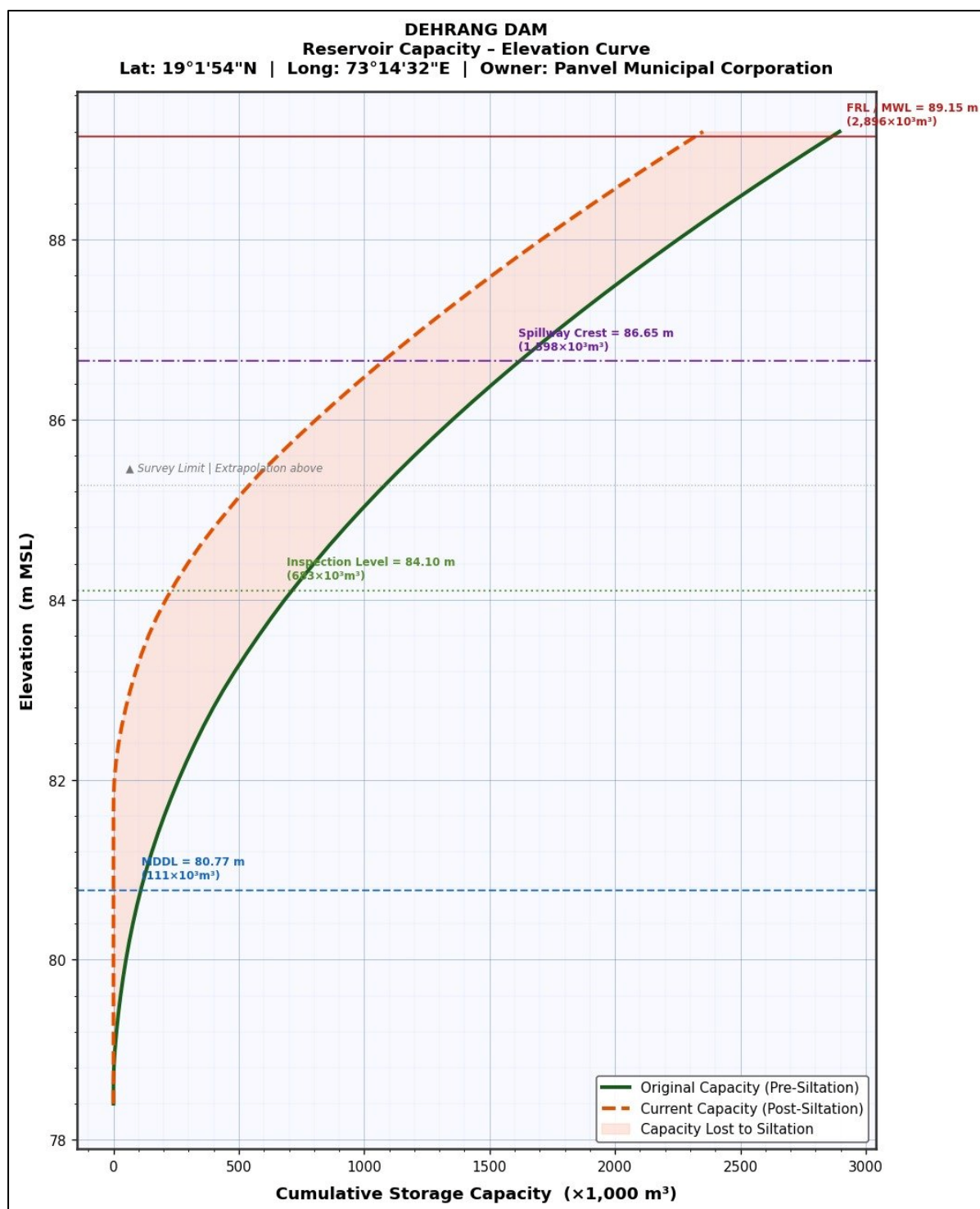


Figure A1: Dehrang Dam – Reservoir Elevation-Area-Capacity Curve (Pre- and Post-Siltation)

The Elevation-Area-Capacity (EAC) relationship for Dehrang Reservoir was derived from a detailed pre-monsoon sludge survey conducted in April 2025 by Yash Engineering Consultant Pvt. Ltd. using a systematic 10 m x 10 m grid-based bathymetric survey method (box method). Siltation volumes were quantified by comparing the 2025 survey against the pre-siltation EAC curve from the original dam

design records. Volumes were computed using the Trapezoidal Integration (prismoidal) method between successive elevation contours.

The survey confirms a total siltation volume of $544.14 \times 10^3 \text{ m}^3$ accumulated primarily below the spillway gate sill level (86.65 m MSL), with the top of silt layer identified at approximately El. 85.40 m MSL. The gross storage at FRL (89.15 m MSL) is $2,351.58 \times 10^3 \text{ m}^3$ (2.352 MCM), representing an 18.8% reduction from the original design capacity and a significant reduction from the post-gate-installation revised capacity of 3.507 MCum. The complete EAC table is presented below; this data was entered directly into the HEC-RAS Storage Area element for reservoir drawdown simulation.

Sl.	Elev. (m MSL)	Orig. Area (ha)	Incr. Vol. (10^3 m^3)	Cum. Orig. (10^3 m^3)	Curr. Area (ha)	Incr. Vol. (10^3 m^3)	Cum. Curr. (10^3 m^3)	Silt Vol. (10^3 m^3)	Silt %	Key Level
78.40	0	0	0	0	0	0	0	0	100	
79.00	2.68	4.28	5.98	0	0	0	5.98	2.68	100	
80.00	6.11	11.43	49.91	0	0	0	49.91	6.11	100	MDDL approx.
81.00	10.61	19.97	131.25	0	0	0	131.25	10.61	100	
82.00	15.18	29.37	259.80	2.55	3.05	3.60	256.20	12.63	98.6	
83.00	21.36	41.07	439.17	9.72	18.33	67.64	371.52	11.65	84.6	
84.00	27.58	53.40	683.36	19.54	36.81	208.85	474.51	8.04	69.4	
85.00	33.44	65.70	988.90	29.45	58.77	459.03	529.87	3.99	53.6	
85.40	35.77	70.38	1127.33	35.77	65.24	583.19	544.14	0	48.3	Silt top
86.00	39.25	77.34	1352.41	39.25	77.34	808.27	544.14	0	40.2	
86.65	--	--	1588	--	--	1044	544.14	0	34.3	Spillway Crest
87.00	44.93	88.73	1773.39	44.93	88.73	1229.25	544.14	0	30.7	
88.00	50.49	99.88	2250.60	50.49	99.88	1706.46	544.14	0	24.2	
89.00	55.93	110.78	2782.79	55.93	110.78	2238.65	544.14	0	19.6	
89.15	57.00	112.93	2895.72	57.00	112.93	2351.58	544.14	0	18.8	FRL

Annexure I: Dehrang Reservoir Elevation-Area-Capacity Table (2025 Pre-Monsoon Survey)

Note: Key elevations: FRL/MWL = 89.15 m MSL (gross storage $2,351.58 \times 10^3 \text{ m}^3$). Spillway Gate Sill = 86.65 m MSL. MDDL = 80.77 m MSL (approx.). RBL = 75.40 m MSL. Top of silt layer = 85.40 m MSL. Total siltation volume = $544.14 \times 10^3 \text{ m}^3$ (18.8% of FRL capacity). Data source: Yash Engineering Consultant Pvt. Ltd., April 2025

Annexure II - IDF Computation Tables

A. Annual Maximum Daily Rainfall Series - Panvel IMD Station (1992-2024)

The annual maximum daily rainfall series for Panvel IMD Station (Station ID: 102187319, 18.9833 N, 73.1167 E) was compiled from 31 years of records (1992-2024) with 2 missing years (1997, 2006). The series is ranked in descending order and empirical return periods assigned using the Weibull plotting position formula $F = m/(N+1)$, where m is rank and $N = 31$. The 2005 event (473.5 mm) was flagged as a statistical outlier by Grubbs test ($G = 3.46 > G_{crit} = 2.93$) but retained conservatively per CWC (1989) guidance.

Sr.	Year	Annual Max (mm)	Rank	Weibull F	Emp. T (yr)
1	2005	473.5 OUTLIER	1	0.031	32.0
2	2011	315.0	2	0.063	16.0
3	2003	314.0	3	0.094	10.67
4	2020	306.8	4	0.125	8.0
5	2023	290.0	5	0.156	6.4
6	2021	284.2	6	0.188	5.33
7	2001	276.0	7	0.219	4.57
8	2019	271.8	8	0.250	4.0
9	2013	256.0	9	0.281	3.56
10	2017	254.0	10	0.313	3.2
11	2000	249.0	11	0.344	2.91
12	2007	245.0	12	0.375	2.67
13	1998	241.6	13	0.406	2.46
14	2009	231.6	14	0.438	2.29
15	2004	216.4	15	0.469	2.13
16	1996	209.2	16	0.500	2.0
17	1994	207.6	17	0.531	1.88
18	1992	207.4	18	0.563	1.78
19	2016	193.2	19	0.594	1.68
20	2002	189.0	20	0.625	1.6
21	2018	180.0	21	0.656	1.52
22	2008	176.2	22	0.688	1.45
23	2014	172.6	23	0.719	1.39
24	2022	172.2	24	0.750	1.33
25	2012	163.0	25	0.781	1.28
26	2010	162.8	26	0.813	1.23
27	2024	155.0	27	0.844	1.19

Sr.	Year	Annual Max (mm)	Rank	Weibull F	Emp. T (yr)
28	2015	152.0	28	0.875	1.14
29	1999	150.2	29	0.906	1.10
30	1993	143.0	30	0.938	1.07
31	1995	118.6	31	0.969	1.03

Annexure II-A: Annual Maximum Daily Rainfall Series - Panvel Station 1992-2024 (Ranked Descending)

B. Gumbel EV-I Frequency Analysis - Computation Table

Gumbel EV-I design rainfalls were computed using the Chow frequency factor formula: $XT = \bar{x} + K \cdot S$, where $K = (y_T - y_n)/S_n$. Statistical parameters: $\bar{x} = 225.06$ mm; $S = 71.84$ mm; $y_n = 0.5370$ ($N=31$); $S_n = 1.1156$ ($N=31$). Both Gumbel EV-I and Lognormal distributions passed K-S and Chi-square goodness-of-fit tests at the 5% significance level (Annexure II-C).

T (yr)	1/T	F=1-1/T	Reduced Variate y_T	K Factor	XT (mm)	95% CL Lower	95% CL Upper
2	0.500	0.500	0.3665	-0.1528	214.08	193.16	235.00
5	0.200	0.800	1.4999	0.8631	287.07	249.12	325.03
10	0.100	0.900	2.2504	1.5358	335.40	282.99	387.81
25	0.040	0.960	3.1985	2.3857	396.46	324.84	468.07
50	0.020	0.980	3.9019	3.0162	441.76	355.60	527.92
100	0.010	0.990	4.6001	3.6420	486.72	386.00	587.45
200	0.005	0.995	5.2958	4.2656	531.52	416.20	646.84
1000	0.001	0.999	6.9073	5.7100	635.29	486.00	784.59

Annexure II-B: Gumbel EV-I Design Rainfall Computation - All Return Periods

C. Statistical Parameters and Goodness-of-Fit Summary

Parameter	Value
N (Record Length)	31 years (1992-2024)
Missing Years	1997 and 2006
Sample Mean ($\bar{x}$)	225.06 mm
Sample Std Dev (S)	71.84 mm
Coefficient of Variation (Cv)	0.319
Skewness Coefficient (Cs)	1.357
Gumbel y_n (N=31)	0.5370
Gumbel S_n (N=31)	1.1156
Lognormal u_Y	5.3720
Lognormal s_Y	0.2983
K-S Test - Gumbel EV-I	PASS ($D=0.0722 < D_{crit}=0.2443$)

Parameter	Value
K-S Test - Lognormal	PASS ($D=0.0695 < D_{crit}=0.2443$)
Chi-Square Test (Gumbel)	PASS ($\chi^2=0.129 < 5.991$)
2005 Outlier Status	CONFIRMED ($G=3.46 > G_{crit}=2.93$; retained conservatively)

Annexure II-C: Statistical Parameters and Goodness-of-Fit Test Results

Annexure III - Alternating Block Method (ABM) Storm Distribution (Reference)

The 100-year 24-hour design storm temporal distribution was computed using the Alternating Block Method (ABM) as a reference calculation to verify storm structure and provide a cross-check with the SCS Type II distribution used in HEC-HMS. The ABM (Chow, Maidment & Mays, 1988) arranges incremental rainfall depths derived from the IDF curves in an alternating pattern centred on Hour 12, with the largest block placed at the storm centre.

Total storm depth = $P_{24,100} = 486.72$ mm (Gumbel EV-I, 100-year return period). The peak block at Hour 12 = 168.74 mm = 34.67% of total depth. The ABM distribution is provided for reference and comparison only.

Hour (t)	Incremental Rainfall (mm)	Intensity (mm/hr)	Cumulative Rainfall (mm)	% of Total Rainfall	Rank of Block
1	7.06	7.06	7.06	1.45	23
2	7.51	7.51	14.57	1.54	21
3	8.04	8.04	22.61	1.65	19
4	8.68	8.68	31.29	1.78	17
5	9.46	9.46	40.75	1.94	15
6	10.45	10.45	51.20	2.15	13
7	11.73	11.73	62.93	2.41	11
8	13.51	13.51	76.44	2.78	9
9	16.17	16.17	92.61	3.32	7
10	20.68	20.68	113.29	4.25	5
11	30.77	30.77	144.06	6.32	3
12	168.74	168.74	312.80	34.67	1 <- PEAK
13	43.86	43.86	356.66	9.01	2
14	24.49	24.49	381.15	5.03	4
15	18.08	18.08	399.23	3.71	6
16	14.69	14.69	413.92	3.02	8
17	12.55	12.55	426.47	2.58	10
18	11.04	11.04	437.51	2.27	12
19	9.92	9.92	447.43	2.04	14
20	9.05	9.05	456.48	1.86	16
21	8.35	8.35	464.83	1.71	18
22	7.76	7.76	472.59	1.60	20
23	7.28	7.28	479.86	1.49	22
24	6.86	6.86	486.72	1.41	24

Annexure III: ABM Design Storm Temporal Distribution - Reference Only ($P_{24,100} = 486.72$ mm)

Note: Sum of all incremental blocks = 486.72 mm (CHECK PASS). This ABM hyetograph was NOT used as the rainfall input to HEC-HMS. HEC-HMS used the Hypothetical Storm with SCS Type II 24-hour distribution, which is the appropriate paired distribution for the SCS Unit Hydrograph transform method. The ABM is provided here for completeness and reference verification.

Annexure IV - Froehlich (2008) Breach Parameter Derivation

Breach parameters for both failure scenarios were derived from Froehlich (2008) empirical equations. The methodology is summarised below for transparency and QA purposes. The Froehlich (2008) equations are based on regression analysis of 74 documented dam failure case studies and are recommended by CWC (2016) EAP Guidelines and CDSO (2018) for breach parameter estimation in Indian dam safety analysis.

A. Input Parameters

Reservoir volume at trigger WSL:

- Overtopping (WSL = 90.0 m MSL): $V_w = 2,351,580 \text{ m}^3$ (gross storage at FRL 89.15 m MSL, conservative estimate used)
- Piping (WSL = 89.15 m MSL): $V_w = 2,351,580 \text{ m}^3$ (gross storage from 2025 survey)

Breach height H_b :

- Overtopping: $H_b = \text{Trigger WSL} - \text{Final Breach Bottom Elev} = 90.0 - 81.4 = 8.6 \text{ m}$
- Piping: $H_b = \text{FRL} - \text{RBL} = 89.15 - 75.40 = 13.75 \text{ m}$

B. Froehlich (2008) Equations and Computed Values

$$\text{Average Breach Width: } B_{avg} = 0.1803 * K_0 * V_w^{0.32} * H_b^{0.19}$$

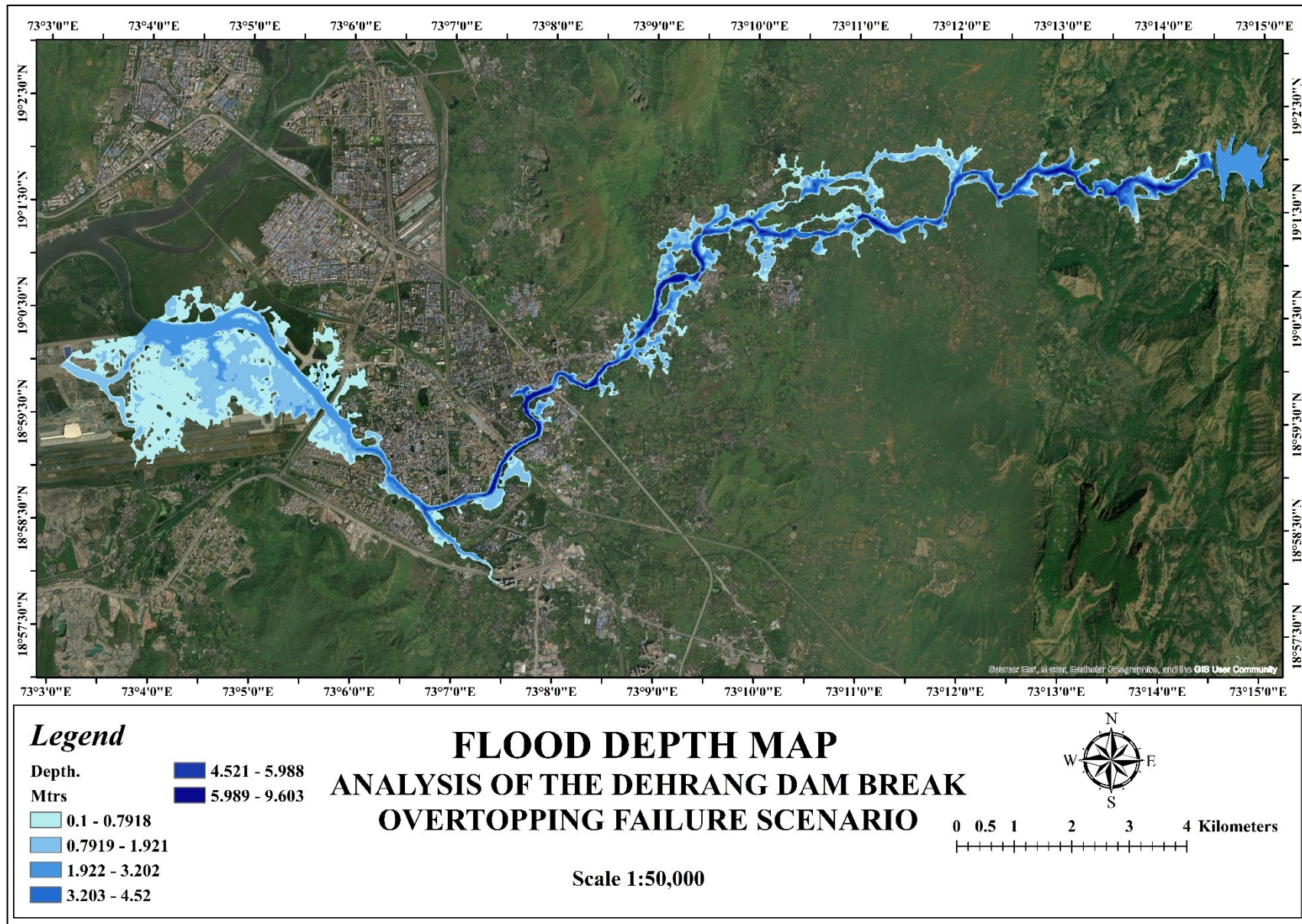
$$\text{Formation Time: } t_f = 0.0179 * V_w^{0.53} * H_b^{(-0.90)}$$

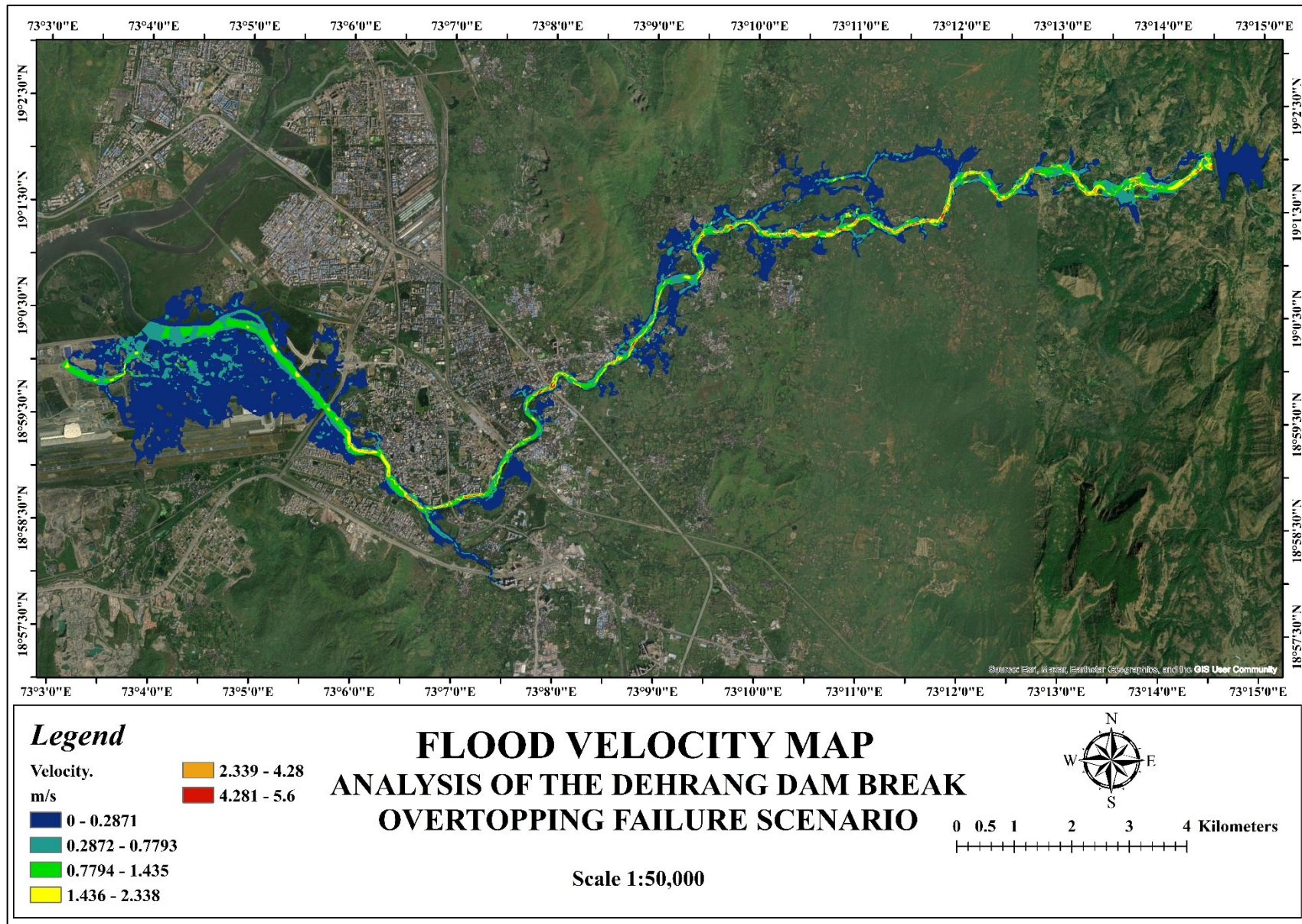
$$K_0 = 1.0 \text{ (overtopping); } K_0 = 0.7 \text{ (piping); } V_w \text{ in } m^3; H_b \text{ in } m$$

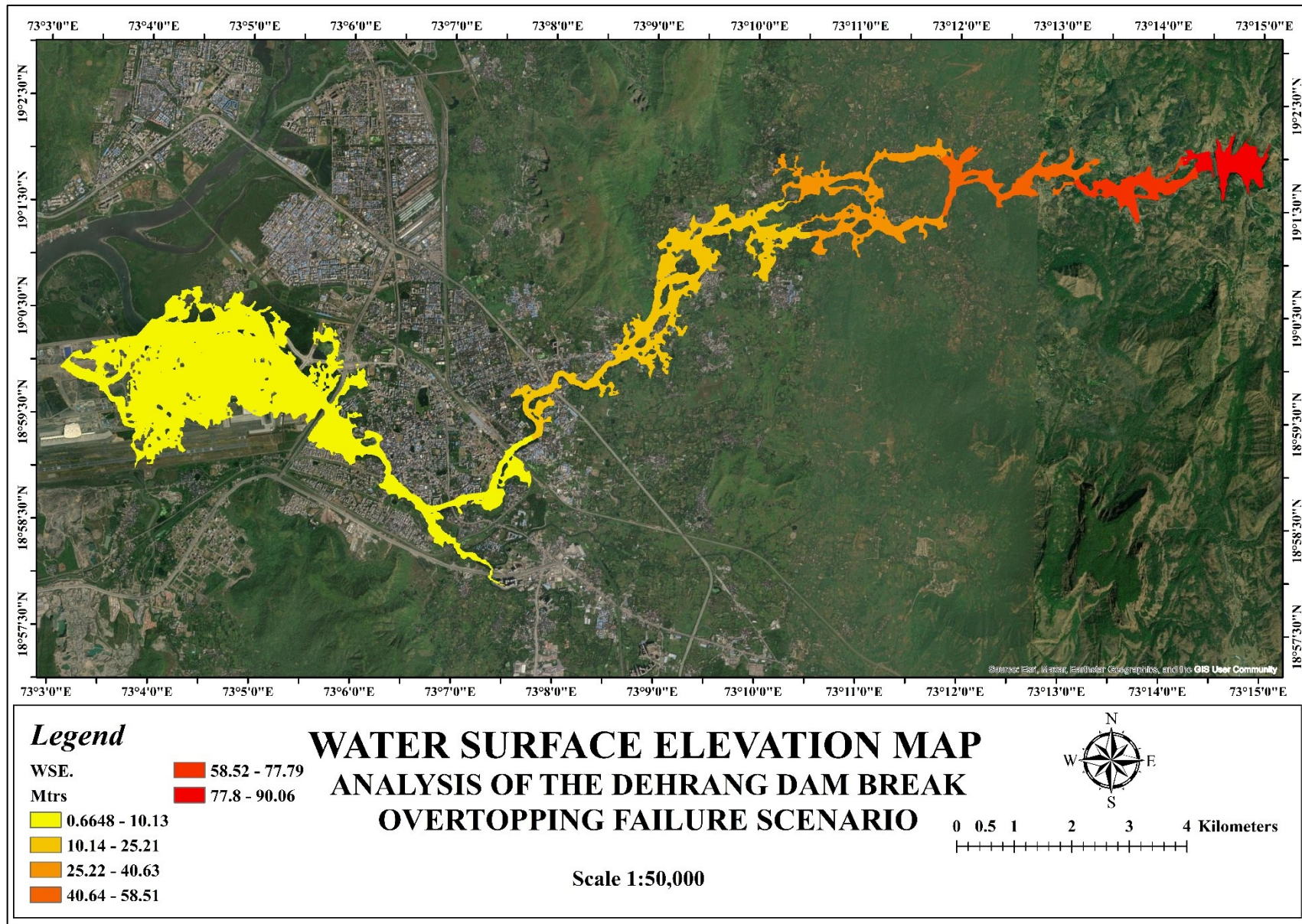
Parameter	Overtopping (OT)	Piping (PI)
Failure mode constant K_0	1.0	0.7
Volume of water V_w (m^3)	2,538,650 (at pool elevation WSL 90.0 m MSL, from EAC)	2,238,652 (at pool elevation WSL 89.0 m MSL, from EAC)
Breach height H_b (m)	8.6 (90.0 - 81.4 m final breach invert, overtopping)	7.75 (89.15 - 81.4 m final breach invert)
Froehlich B_{avg} (m)	32 m (model uses 34 m)	33 m
Froehlich t_f (hrs)	1.0 hr (model uses 1.04 hrs)	0.98 hrs
Froehlich side slope z (H:V)	1.0 (overtopping)	1.0 (piping, per Froehlich 2008 for this case)
Final bottom width (model)	34 m	23 m
Final bottom elevation (model)	81.4 m MSL	81.4 m MSL
Breach weir coefficient	1.44	-
Piping coefficient	-	0.5

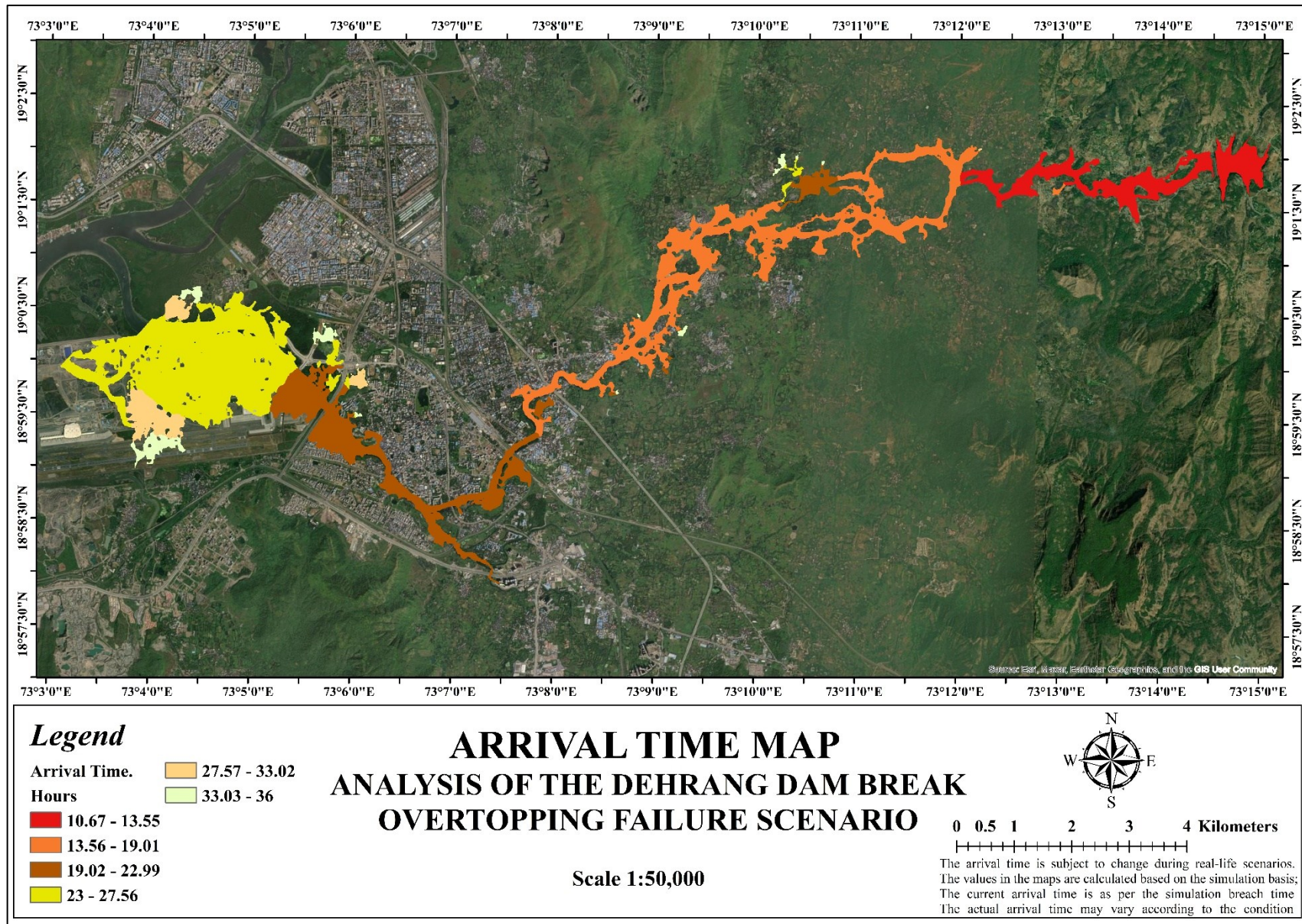
Annexure IV: Froehlich (2008) Breach Parameter Derivation Summary

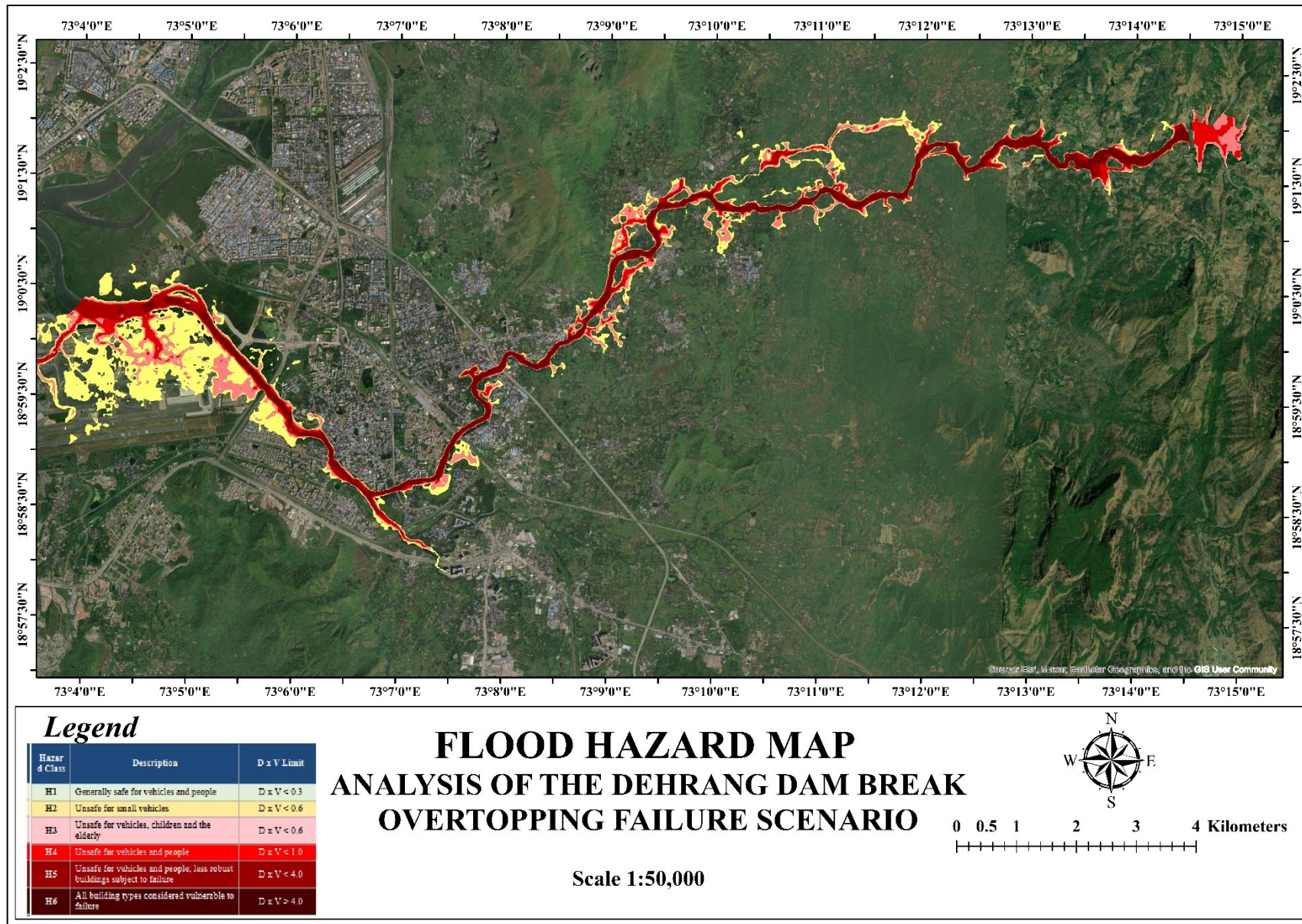
Note: The overtopping model breach parameters (34 m bottom width, 1.04 hr formation time, 81.4 m invert) are consistent with Froehlich (2008) equations and were entered as User Entered Data in HEC-RAS Plan DEH-OVTP-DBA. The piping scenario parameters (33 m bottom width, 0.98 hr formation time, 81.4 m invert, Initial Piping Elevation 81.5 m MSL) were derived from Froehlich (2008) using pool volume $V_w = 2,238,652 \text{ m}^3$ and breach height $H_b = 7.75 \text{ m}$, and entered as User Entered Data in HEC-RAS Plan DEH-PIPG-DBA

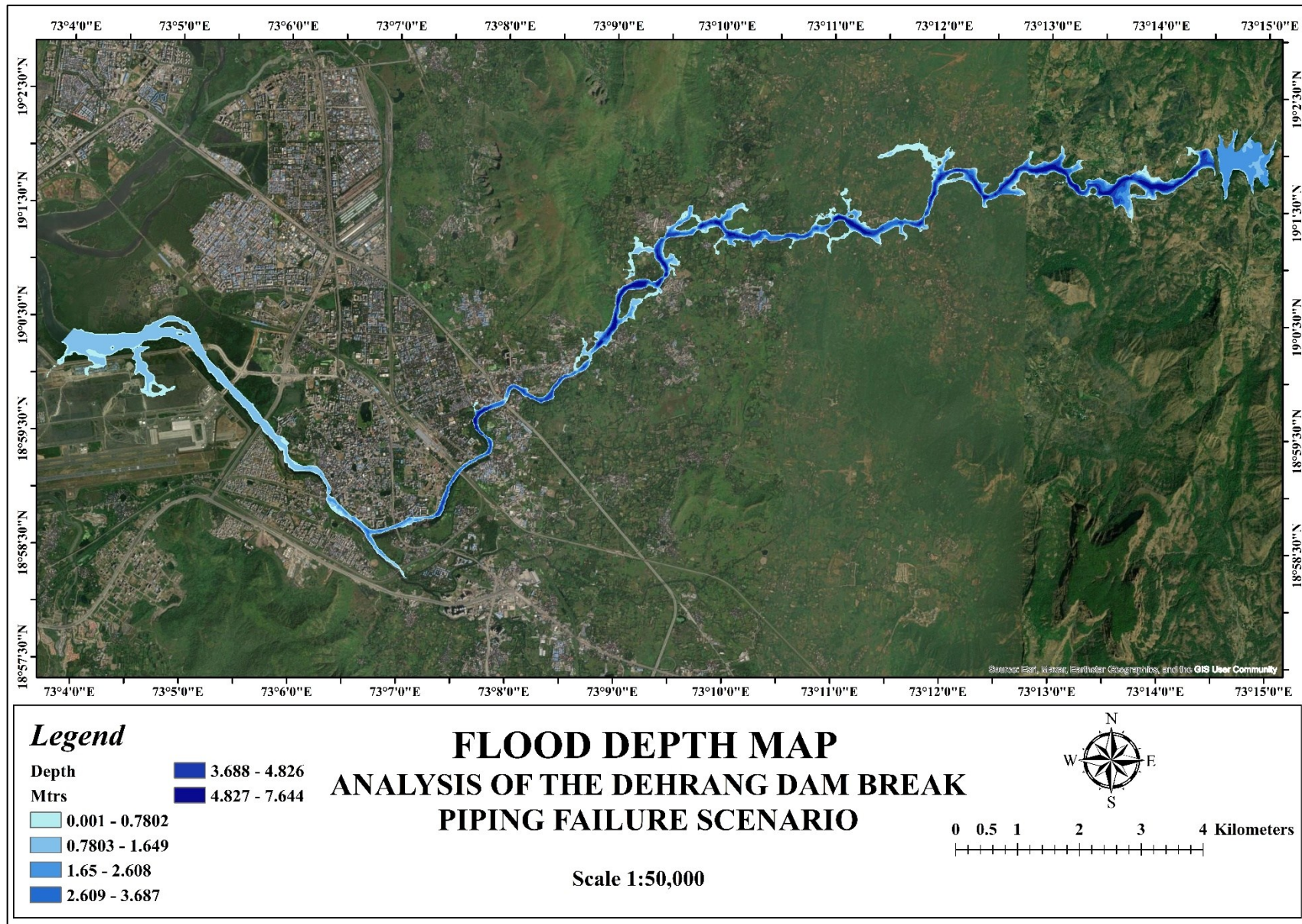


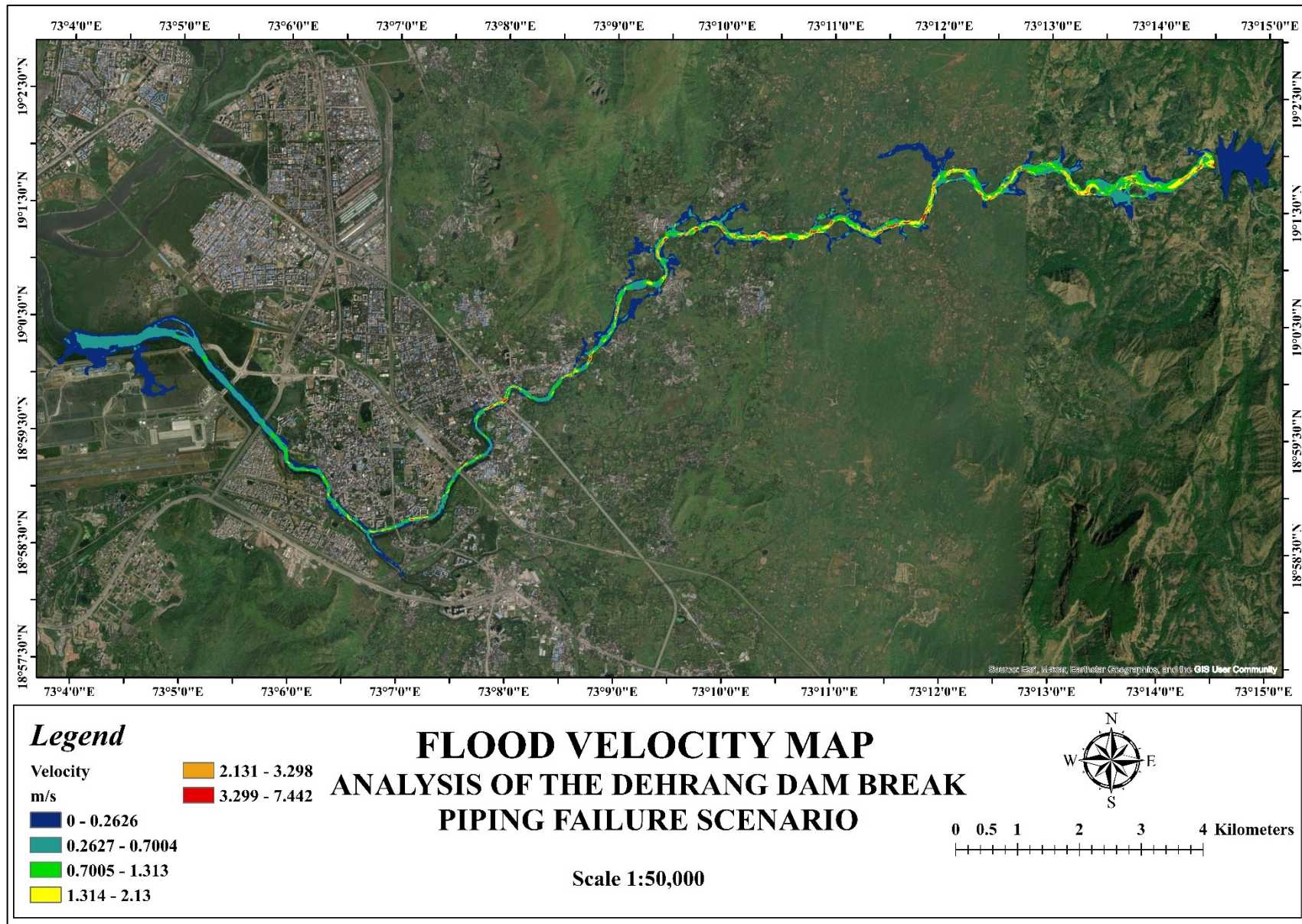


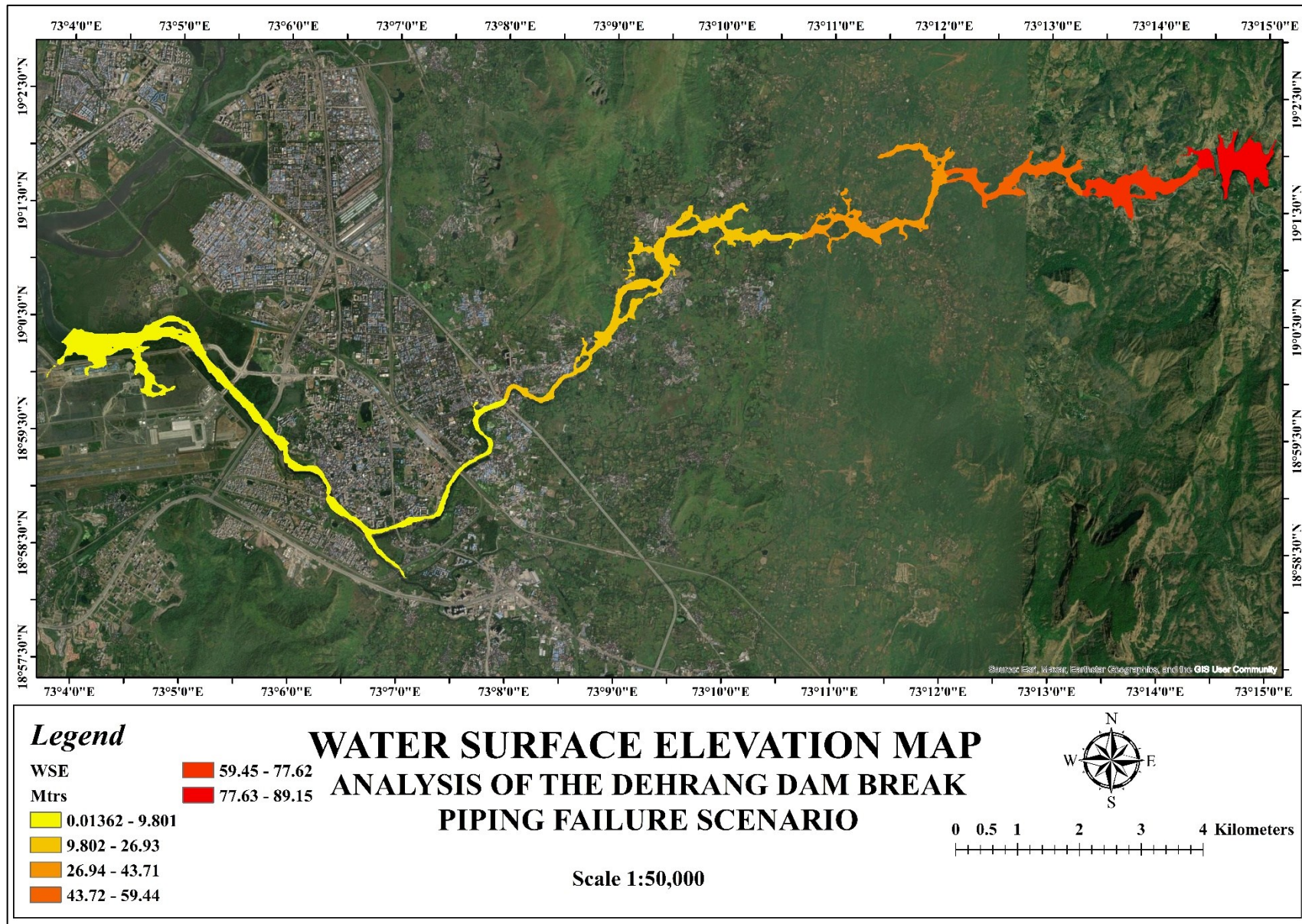


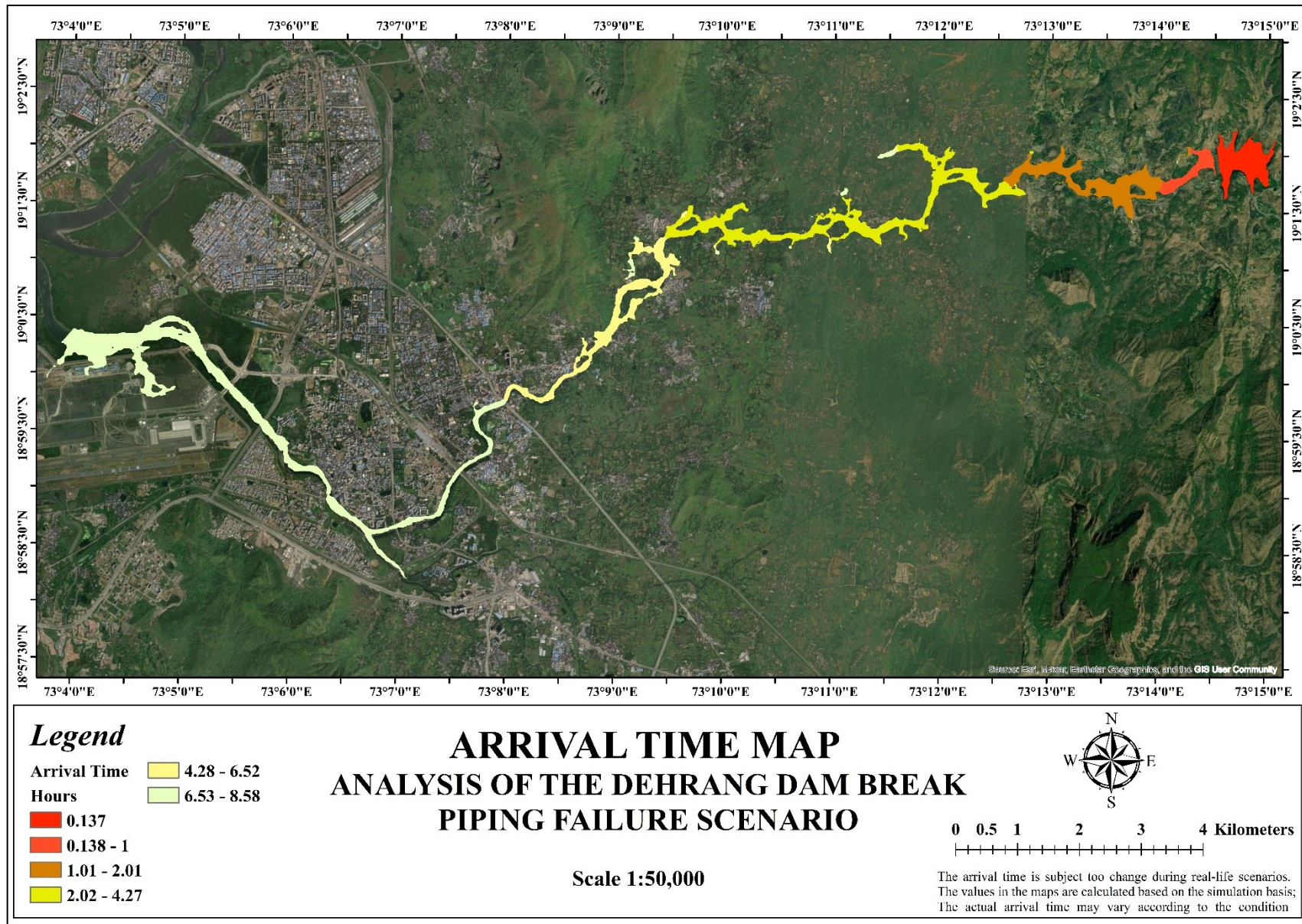


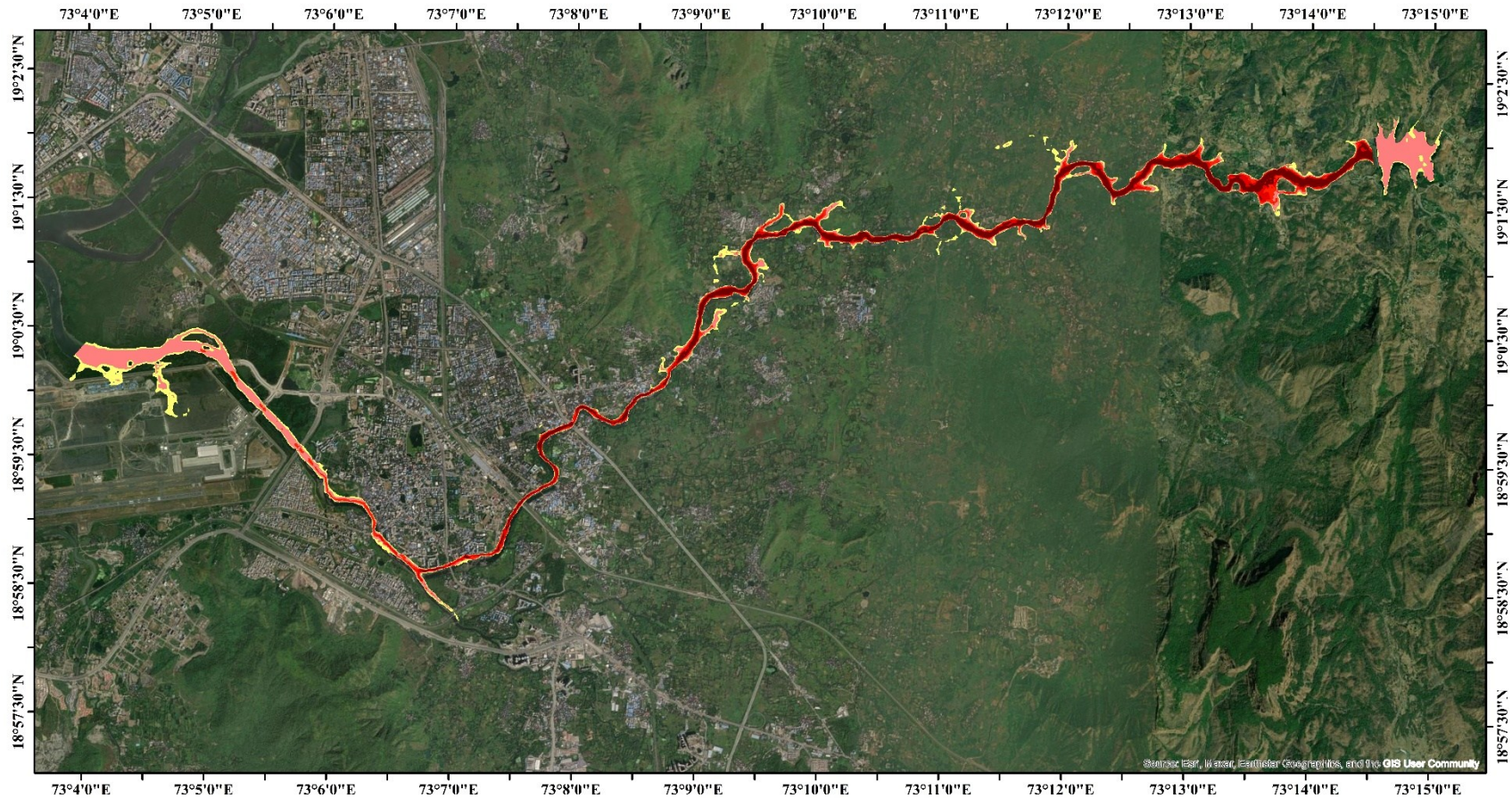










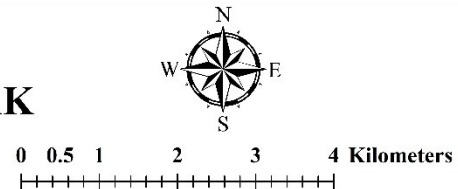


Legend

Hazard Class	Description	D x V Limit
H1	Generally safe for vehicles and people	$D \times V < 0.3$
H2	Unsafe for small vehicles	$D \times V < 0.6$
H3	Unsafe for vehicles, children and the elderly	$D \times V < 0.6$
H4	Unsafe for vehicles and people	$D \times V < 1.0$
H5	Unsafe for vehicles and people; less robust buildings subject to failure	$D \times V < 4.0$
H6	All building types considered vulnerable to failure	$D \times V > 4.0$

FLOOD HAZARD MAP ANALYSIS OF THE DEHRANG DAM BREAK PIPING FAILURE SCENARIO

Scale 1:50,000



Note: The H1 hazard class is not severe and has very low pixel visibility because the depth values range from 0.01 to 0.05 meters. Therefore, the Hazard Vulnerability Index should be used only for preliminary planning, as the actual hazard may vary with terrain and other factors.